

Response Loops and Constitutive Rigidity in Operational Lorentz Geometry: Torsion, Simple Bivectors, and the Einstein–Cartan Sector

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Abstract We investigate the first-order gravitational information carried by the operational generator–displacement response underlying the motion–staticity reconstruction of Lorentzian geometry. The starting point is a faithful local response

$$\Theta_x(P, \xi) = P(e_x \xi),$$

where P is an internal translation-generator covector, $\xi \in T_x M$ is an infinitesimal spacetime displacement, and $e : TM \rightarrow V$ is the solder map reconstructed from the response. Rather than imposing vanishing torsion as a property of the connection, we ask whether torsion can itself be characterized operationally.

For a closed infinitesimal path $\gamma = \partial \Sigma$, we define a covariant response phase by transporting the generator P with the Lorentz connection and integrating the local soldering response along the path. For a loop of linear size ℓ , the leading small-loop expansion is

$$\log W_\gamma(P) = -\frac{i}{\hbar} \int_\Sigma P_a T^a + O(\ell^3),$$

where

$$T^a = de^a + \omega^a_b \wedge e^b$$

is the torsion two-form. Since the positive massive generator shell spans the internal dual space, vanishing of the area-order response phase for every shell generator and every sufficiently small contractible loop implies

$$T^a = 0.$$

Thus torsion-free geometry may be formulated, within the operational framework, as a closed-response condition rather than as a primitive connection assumption.

The same local response determines internal edge vectors $E_\xi = e(\xi)$ and hence area bivectors

$$B^{ab}(\xi, \zeta) = E_\xi^{[a} E_\zeta^{b]}, \quad B^{ab} = e^a \wedge e^b.$$

Their simplicity is therefore inherited directly from operational soldering. We then separate this kinematical area structure from the gravitational constitutive relation. If the excitation associated with curvature is related locally and linearly to B by a Lorentz-equivariant map

$$H = \mathfrak{C}(B), \quad \mathfrak{C} : \Lambda^2 V \rightarrow \Lambda^2 V,$$

then in oriented four-dimensional Lorentzian geometry

$$\mathfrak{C} = \alpha I + \beta \star .$$

Consequently, within the local first-order sector linear in curvature and containing no additional gravitational fields, the most general such response consists of the Palatini–Einstein–Cartan and Holst directions.

In the spinless torsion-free branch the identity/Holst contribution does not modify the classical bulk field equations, while the Hodge-dual component supplies the Einstein–Cartan tensor structure. Its overall magnitude, however, remains arbitrary. The operational state, soldering, loop-response, and Lorentz-covariance assumptions therefore determine neither Newton’s constant nor the relative Holst/Immirzi coefficient. We formulate this result as a constitutive-underdetermination theorem: the gravitational coupling is a response stiffness relating operational area geometry to curvature excitation, rather than a parameter fixed by the compact motion–staticity kinematics.

When intrinsic spin is admitted, the connection equation takes the schematic form

$$DH_{ab} = \tau_{ab},$$

and nonzero spin current generates torsion. The corresponding torsion is then directly visible in the infinitesimal closed-response phase. The ratio α/β , which is invisible in the torsion-free spinless bulk theory, can consequently enter the spin–torsion response sector.

Finally, we study an optional constitutive reciprocity principle. If the graph of $H = \mathfrak{C}B$ is invariant under the exchange

$$(B, H) \mapsto (\zeta H, -\zeta^{-1} B),$$

then

$$\mathfrak{C}^2 = -\zeta^{-2} I.$$

For real Lorentz-equivariant constitutive maps this condition selects

$$\mathfrak{C} = \pm \zeta^{-1} \star .$$

This selects the Einstein–Cartan direction up to an overall scale and sign, but the reciprocity condition is an additional constitutive postulate rather than a consequence of the underlying operational response. The work therefore identifies both the first-order geometric information encoded by operational response loops and the precise constitutive boundary beyond which additional gravitational physics must be supplied.

Keywords: operational soldering; torsion; response loops; Einstein–Cartan gravity; constitutive gravity; Holst term; Immirzi parameter; simple bivectors; Lorentz covariance; Hodge duality; motion–staticity geometry

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1 Introduction

The reconstruction of spacetime geometry from operational data raises two logically different questions. The first is kinematical: how does one obtain a Lorentzian tangent geometry from quantities that are directly associated with the response of physical generators to spacetime displacements? The second is dynamical: once such a geometry has been reconstructed, what determines the law by which it responds to matter and curvature?

The motion–staticity (MS) programme developed in the preceding works addresses the first question by starting from a compact observer-dependent state representation rather than from a preassigned spacetime metric. In its massive classical realization, the normalized state variables satisfy

$$|\mathbf{D}|^2 + S^2 = 1, \quad S > 0, \quad (1)$$

with

$$\mathbf{D} = \frac{\mathbf{v}}{c}, \quad S = \frac{d\tau}{dt}. \quad (2)$$

The projective map

$$\Phi(\mathbf{D}, S) = \frac{1}{S}(1, \mathbf{D}) \quad (3)$$

identifies the open compact hemisphere with the future unit Lorentz hyperboloid. In the flat massive sector, consistency between this state representation and Hamiltonian dynamics selects the ordinary relativistic mass shell under the stated regularity assumptions, while composition-compatible observer covariance recovers the proper orthochronous Lorentz action.

The curved-spacetime reconstruction introduces an additional operational datum. Let V_x denote the local internal Lorentz vector space and let

$$\Sigma_{Q,x}^+ \subset V_x^*$$

be a positive massive generator shell. For

$$P \in \Sigma_{Q,x}^+, \quad \xi \in T_x M,$$

the scalar

$$\Theta_x(P, \xi) \quad (4)$$

represents the infinitesimal generator–displacement response. Under additivity, continuity, smoothness, and operational faithfulness, the response determines a unique nondegenerate solder map

$$e_x : T_x M \xrightarrow{\sim} V_x \quad (5)$$

such that

$$\boxed{\Theta_x(P, \xi) = P(e_x \xi)} \quad (6)$$

The internal Lorentz form η then induces the spacetime metric

$$\boxed{g = e^* \eta} \quad (7)$$

In this sense Lorentzian tangent geometry records how translation generators respond to infinitesimal spacetime displacements rather than being introduced as an independent primitive structure.

The preceding quantum-response analysis also shows that preserving the local additive response law exactly in a unitary representation does not by itself produce a noncommutative quantum geometry. Exact generator and displacement composition force the local response algebra to be commutative. The present work addresses a different question. We retain the local operational response but allow it to be compared around closed spacetime paths. The resulting problem is no longer determined by the algebra at a single spacetime point: it depends on the connection relating local Lorentz frames at neighboring events.

Let

$$\omega^a_b \quad (8)$$

be a metric-compatible Lorentz connection. The associated curvature and torsion are

$$R^a_b = d\omega^a_b + \omega^a_c \wedge \omega^c_b, \quad (9)$$

$$T^a = de^a + \omega^a_b \wedge e^b. \quad (10)$$

The geometrical interpretation of torsion as translational nonclosure is classical and well established [?, ?]. The objective here is therefore not to introduce a new definition of torsion. Instead, we ask whether the particular generator–displacement response (6) provides a direct operational realization of this translational nonclosure.

Consider a sufficiently small closed path

$$\gamma = \partial\Sigma$$

based at x , and transport a generator covector P_a along γ using the Lorentz connection. We define the covariant closed-response phase

$$\boxed{W_\gamma(P) := \exp \left[-\frac{i}{\hbar} \oint_\gamma P_a e^a \right]} \quad (11)$$

where P is understood to be parallel transported along the loop. For a loop of characteristic size ℓ , the leading covariant small-loop expansion gives

$$\boxed{\log W_\gamma(P) = -\frac{i}{\hbar} \int_\Sigma P_a T^a + O(\ell^3)} \quad (12)$$

The leading area-order phase therefore probes the contraction of torsion with the transported translation generator.

This observation leads to the first principal result of the paper. Because a positive massive generator shell spans V_x^* , if

$$W_\gamma(P) = 1 + O(\ell^3) \quad (13)$$

for every sufficiently small contractible loop and every generator in the shell, then

$$\boxed{T^a = 0} \quad (14)$$

Conversely, vanishing torsion removes the area-order translational response phase. Within the operational reconstruction, the torsion-free condition can therefore be replaced by a *response-loop criterion*. This does not make zero torsion a consequence of the compact MS state alone. It provides an operational statement equivalent to zero torsion once soldering and Lorentz transport have been established.

The response also contains an area structure. For $\xi \in T_x M$, define the internal edge vector

$$E_\xi^a := e^a(\xi). \quad (15)$$

Since

$$\Theta(P, \xi) = P_a E_\xi^a, \quad (16)$$

the vector E_ξ is recoverable from the generator dependence of the response. For two infinitesimal displacements ξ, ζ , the corresponding internal area bivector is

$$\boxed{B^{ab}(\xi, \zeta) = E_\xi^{[a} E_\zeta^{b]}} \quad (17)$$

Equivalently,

$$\boxed{B^{ab} = e^a \wedge e^b} \quad (18)$$

The bivector is therefore simple by construction. Area bivectors sharing an operational edge satisfy the corresponding cross-simplicity relations.

Simplicity constraints and the representation $B = e \wedge e$ are standard ingredients of Plebanski and BF formulations of general relativity and their spin-foam descendants [?, ?, ?]. No new simplicity constraint is claimed here. The distinction is one of derivational order: in the present construction the simple bivectors are not introduced as independent BF variables subsequently constrained to be geometric; they arise as composite area responses once the operational solder map has been reconstructed.

At this point an important logical boundary appears. The response determines the solder form and therefore the simple area bivector B , but it does not determine how this area variable couples to Lorentz curvature. That relation belongs to a separate *constitutive sector*.

Let

$$H_{ab} = \mathfrak{C}(B)_{ab} \quad (19)$$

denote the gravitational excitation conjugate to the Lorentz curvature, and suppose that the constitutive map

$$\mathfrak{C} : \Lambda^2 V \longrightarrow \Lambda^2 V \quad (20)$$

is local, real, linear, and equivariant under $SO^+(1, 3)$. In oriented four-dimensional Lorentzian geometry the commutant of the Lorentz action on real bivectors is generated by the identity and the internal Hodge operator. Hence

$$\boxed{\mathfrak{C} = \alpha I + \beta \star} \quad (21)$$

Here

$$\star^2 = -I \quad (22)$$

on internal bivectors.

Equation (21) is the second principal structural result used in this work. The representation-theoretic fact itself is standard. Its role here is to isolate the complete local linear Lorentz-equivariant constitutive freedom remaining after the operational geometry has already fixed $B = e \wedge e$.

A first-order curvature action constructed from this data consequently has, up to normalization and the cosmological volume term, the form

$$S_g = \frac{1}{2} \int B^{ab} \wedge (\alpha R_{ab} + \beta (\star R)_{ab}) + S_\Lambda. \quad (23)$$

Equivalently one may parameterize the nondegenerate Einstein–Holst family by

$$\mathfrak{C} = \frac{1}{2\kappa} \left(\star + \frac{1}{\gamma} I \right), \quad (24)$$

up to sign and normalization conventions.

The appearance of the Einstein–Cartan and Holst tensor structures is not itself new [?, ?]. Likewise, constitutive formulations of gravity are well established, particularly in premetric and teleparallel approaches [?, ?]. The present question is instead what part of (24) follows from the operational response and what part must remain independent physical input.

In the spinless torsion-free sector, the answer is especially sharp. The Nieh–Yan identity

$$d(e_a \wedge T^a) = T_a \wedge T^a - e_a \wedge e_b \wedge R^{ab} \quad (25)$$

implies that

$$T^a = 0 \implies e_a \wedge e_b \wedge R^{ab} = 0. \quad (26)$$

Thus the identity/Holst direction does not modify the classical torsion-free bulk equations, whereas the Hodge-dual direction produces the ordinary Palatini–Einstein–Cartan equations. Consequently a continuous constitutive family is compatible with exactly the same operational soldering and spinless classical geometry.

This yields a constitutive-underdetermination result:

$$\boxed{\text{operational soldering} + \text{Lorentz covariance} + \text{spinless response-loop closure} \not\Rightarrow (\kappa, \gamma)} \quad (27)$$

In particular, the numerical magnitude of Newton’s constant cannot be deduced from the compact state relation or from the local generator–displacement response. The quantity

$$\frac{1}{\kappa} \sim \frac{c^4}{G} \quad (28)$$

is naturally interpreted here as a gravitational constitutive stiffness: it specifies the magnitude of curvature excitation associated with a given operational area geometry.

The distinction becomes physically relevant when intrinsic spin is included. For first-order matter, the Lorentz connection couples to the spin current

$$\tau_{ab} = 2 \frac{\delta S_m}{\delta \omega^{ab}}. \quad (29)$$

The connection equation takes schematically the form

$$\boxed{DH_{ab} = \tau_{ab}} \quad (30)$$

Since

$$DB^{ab} = T^a \wedge e^b - e^a \wedge T^b, \quad (31)$$

the constitutive relation gives an algebraic spin–torsion response in the nonpropagating connection branch,

$$(\alpha I + \beta \star)(T^a \wedge e^b - e^a \wedge T^b) \sim \tau^{ab}. \quad (32)$$

The torsion generated by spin then enters directly into the closed-response phase (12).

This provides an operational distinction between the overall constitutive magnitude and the relative constitutive orientation. The former is controlled by the gravitational stiffness, whereas the ratio

$$\frac{\alpha}{\beta} \quad (33)$$

is invisible in the torsion-free spinless bulk sector but can enter spin–torsion response. This is closely related to the familiar role of the Holst/Immirzi parameter in first-order gravity with fermionic matter [?, ?, ?]. We do not claim that the operational construction fixes the Immirzi parameter.

The final part of the paper considers whether an additional constitutive principle can reduce the freedom (21). Motivated by reciprocity arguments familiar from premetric electrodynamics [?, ?], consider the area–excitation exchange

$$\boxed{\mathcal{R}_\zeta : (B, H) \mapsto (\zeta H, -\zeta^{-1} B)} \quad (34)$$

where ζ supplies the required constitutive units. If the graph

$$H = \mathfrak{C} B$$

is invariant under $\mathcal{R}_\zeta$, then

$$\boxed{\mathfrak{C}^2 = -\zeta^{-2} I} \quad (35)$$

For

$$\mathfrak{C} = \alpha I + \beta \star, \quad \star^2 = -I,$$

one obtains

$$\mathfrak{C}^2 = (\alpha^2 - \beta^2)I + 2\alpha\beta \star. \quad (36)$$

Pure reciprocity closure therefore implies

$$2\alpha\beta = 0, \quad \alpha^2 - \beta^2 = -\zeta^{-2}. \quad (37)$$

For real coefficients the identity branch is excluded, and hence

$$\boxed{\mathfrak{C} = \pm \zeta^{-1} \star} \quad (38)$$

Thus pure constitutive reciprocity selects the Einstein–Cartan/Hodge-dual direction and removes the Holst component.

The logical status of this conclusion must be kept explicit. Equation (34) is not derived from the compact MS state, from operational soldering, or from local response composition. It is an additional constitutive postulate. Moreover, even when imposed, it fixes only the direction of the constitutive map; the scale ζ^{-1} , and therefore the magnitude associated with Newton’s constant, remains free.

The resulting hierarchy is

$$\begin{aligned}
\Theta &\implies e, g, \\
\text{closed response} &\implies \text{operational torsion diagnostic}, \\
\Theta &\implies B = e \wedge e, \\
\text{Lorentz covariance} &\implies \mathfrak{C} = \alpha I + \beta \star, \\
\text{spinless torsion-free sector} &\implies \text{Einstein classical equivalence class}, \\
\text{pure constitutive reciprocity} &\implies \mathfrak{C} \propto \star \quad (\text{conditionally}), \\
\text{operational principles alone} &\not\Rightarrow G \text{ or } \gamma.
\end{aligned} \tag{39}$$

The purpose of the present work is therefore neither to rediscover Einstein–Cartan gravity nor to derive a numerical gravitational coupling. Its aim is to locate the boundary between operationally reconstructed geometry and gravitational constitutive physics. Torsion is identified with an infinitesimal closed-response defect, simple area bivectors are reconstructed from the same local response, and Lorentz covariance restricts the remaining linear constitutive freedom to a two-dimensional algebra. Spin and reciprocity then probe different parts of that freedom.

The remainder of the paper is organized as follows. Section 2 defines covariant response loops and derives their infinitesimal torsion expansion. Section 3 reconstructs simple area bivectors from the local response. Section 4 classifies local linear Lorentz-equivariant bivector constitutive maps. Section 5 studies the spinless torsion-free sector and proves the constitutive-underdetermination result. Section 6 introduces intrinsic spin and derives the corresponding operational spin–torsion response. Section 7 studies the conditional reciprocity closure principle. Section ?? discusses the relation to Einstein–Cartan, Holst, Plebanski, and premetric constitutive formulations, and isolates the claims that are specific to the operational reconstruction.

2 Covariant Response Loops and Operational Torsion

The local operational response

$$\Theta_x(P, \xi) = P(e_x \xi) \tag{40}$$

relates an internal translation-generator covector $P \in V_x^*$ to an infinitesimal spacetime displacement $\xi \in T_x M$. Once responses at distinct spacetime points are to be compared, however, the internal Lorentz spaces must themselves be related by a connection. The purpose of this section is to show that the corresponding closed-path response contains a direct operational diagnostic of torsion.

Throughout this section, $V \rightarrow M$ denotes an oriented Lorentz vector bundle with internal metric

$$\eta_{ab} = \text{diag}(-1, 1, 1, 1), \tag{41}$$

e^a is a nondegenerate solder form, and

$$\omega^a_b \tag{42}$$

is a smooth metric-compatible Lorentz connection,

$$\omega_{ab} = -\omega_{ba}. \tag{43}$$

Its curvature and torsion are

$$R^a_b = d\omega^a_b + \omega^a_c \wedge \omega^c_b, \tag{44}$$

$$T^a = De^a = de^a + \omega^a_b \wedge e^b. \tag{45}$$

No assumption

$$T^a = 0 \quad (46)$$

is made in the present section.

2.1 Parallel transport of the response generator

Let

$$\gamma : [0, 1] \longrightarrow M, \quad s \longmapsto \gamma(s), \quad (47)$$

be a smooth oriented path with initial point

$$x = \gamma(0). \quad (48)$$

Choose an initial generator

$$P(0) = P \in V_x^*. \quad (49)$$

The generator is transported along γ by the dual Lorentz connection:

$$D_s P_a := \frac{dP_a}{ds} - \dot{\gamma}^\mu \omega_{a\mu}^b P_b = 0 \quad (50)$$

Because the connection is Lorentz compatible,

$$\frac{d}{ds} (\eta^{ab} P_a P_b) = 0. \quad (51)$$

Hence, if the initial generator belongs to a positive massive shell,

$$P \in \Sigma_{Q,x}^+, \quad \eta^{ab} P_a P_b = -Q^2, \quad (52)$$

then its parallel transport remains on the corresponding Lorentz orbit along the path. In particular, Lorentz transport changes the internal representation of the generator but does not change its intrinsic massive norm.

Equation (50) is essential for the closed-response construction. A fixed collection of numerical components P_a would not define a covariant comparison along a path on which the internal Lorentz frame varies.

2.2 Path response and closed-response phase

The infinitesimal response accumulated along γ is defined by

$$I_\gamma(P) := \int_\gamma P_a(s) e^a = \int_0^1 P_a(s) e^a_\mu(\gamma(s)) \dot{\gamma}^\mu(s) ds \quad (53)$$

The corresponding phase response is

$$W_\gamma(P) := \exp \left[-\frac{i}{\hbar} I_\gamma(P) \right] \quad (54)$$

This expression may equivalently be regarded as the continuum limit of the product of local infinitesimal response phases. For a subdivision

$$0 = s_0 < s_1 < \cdots < s_N = 1 \quad (55)$$

and

$$\Delta x_k^\mu = \gamma^\mu(s_{k+1}) - \gamma^\mu(s_k), \quad (56)$$

one has formally

$$W_\gamma(P) = \lim_{N \rightarrow \infty} \prod_{k=0}^{N-1} \exp \left[-\frac{i}{\hbar} \Theta_{\gamma(s_k)}(P(s_k), \Delta x_k) \right]. \quad (57)$$

Since the infinitesimal response entering the present construction is scalar valued, the continuum product reduces to Eq. (54).

Under a local Lorentz transformation

$$e^a \mapsto \Lambda^a_b e^b, \quad P_a \mapsto P_b (\Lambda^{-1})^b_a, \quad (58)$$

with the usual transformation law for ω , the parallel transport equation remains covariant and

$$P_a e^a \quad (59)$$

is invariant. Therefore both

$$I_\gamma(P) \quad (60)$$

and

$$W_\gamma(P) \quad (61)$$

are independent of the internal Lorentz gauge used to evaluate them.

The response is nevertheless conditioned on the base-point generator P . It is therefore not a Wilson-loop trace and should not be identified with an ordinary Lorentz holonomy invariant. The relevant question is instead how the translational response measured by a specified physical generator changes when the associated displacement path closes.

2.3 Infinitesimal loops

Let

$$\gamma_\ell = \partial \Sigma_\ell \quad (62)$$

be a one-parameter family of contractible loops based at x , shrinking smoothly to x as

$$\ell \longrightarrow 0. \quad (63)$$

We assume that the linear size of γ_ℓ is $O(\ell)$, so that

$$\text{length}(\gamma_\ell) = O(\ell), \quad \text{area}(\Sigma_\ell) = O(\ell^2). \quad (64)$$

In a local coordinate chart y^μ centered at x , define the oriented area bivector of the shrinking surface by

$$\Sigma_\ell^{\mu\nu} := \int_{\Sigma_\ell} dy^\mu \wedge dy^\nu = O(\ell^2). \quad (65)$$

For a smooth two-form

$$F = \frac{1}{2} F_{\mu\nu} dy^\mu \wedge dy^\nu, \quad (66)$$

one therefore has

$$\int_{\Sigma_\ell} F = \frac{1}{2} F_{\mu\nu}(x) \Sigma_\ell^{\mu\nu} + O(\ell^3). \quad (67)$$

The leading contribution to a closed response will be of area order $O(\ell^2)$. The main technical issue is to show that the Lorentz holonomy of the transported generator does not generate an independent curvature term at the same order.

2.4 Separation of translational and Lorentz holonomy

We first record the local order estimate underlying the result.

Lemma 2.1 (Parallel-generator estimate). *Let $P(s)$ satisfy Eq. (50) along a shrinking loop γ_ℓ based at x . Then one may choose a smooth local Lorentz gauge satisfying*

$$\omega^a_{b\mu}(x) = 0 \quad (68)$$

such that, uniformly along the shrinking loop for fixed initial P ,

$$\boxed{P_a(s) = P_a(0) + O(\ell^2)} \quad (69)$$

The total Lorentz holonomy of the generator around the loop is also of order

$$P_a(1) - P_a(0) = O(\ell^2). \quad (70)$$

Proof. A smooth Lorentz gauge can always be chosen so that the connection vanishes at one specified point. In such a gauge, smoothness gives

$$\omega^a_{b\mu}(y) = O(|y|) = O(\ell) \quad (71)$$

throughout the shrinking loop.

Writing

$$\gamma_\ell^\mu(s) = x^\mu + O(\ell), \quad (72)$$

one has

$$\dot{\gamma}_\ell^\mu(s) = O(\ell). \quad (73)$$

The parallel-transport equation therefore gives

$$\frac{dP_a}{ds} = \dot{\gamma}^\mu \omega^b_{a\mu} P_b = O(\ell^2), \quad (74)$$

and integration over $0 \leq s \leq 1$ yields Eq. (69).

The nontrivial return holonomy is governed at leading order by the Lorentz curvature,

$$P(1) - P(0) = O(R \ell^2), \quad (75)$$

and hence is likewise $O(\ell^2)$. $\square$

The last statement is important. Curvature does act on the transported generator at area order. However, the closed response

$$I_{\gamma_\ell}(P) = \oint_{\gamma_\ell} P_a(s) e^a \quad (76)$$

contains this generator variation multiplied by a line element of order $O(\ell)$. The resulting correction is therefore $O(\ell^3)$, not $O(\ell^2)$.

More explicitly, Lemma 2.1 implies

$$\begin{aligned} I_{\gamma_\ell}(P) &= \oint_{\gamma_\ell} [P_a(0) + O(\ell^2)] e^a \\ &= P_a(0) \oint_{\gamma_\ell} e^a + O(\ell^3). \end{aligned} \quad (77)$$

This is the step that separates the leading translational nonclosure from the Lorentz holonomy of the generator.

2.5 Infinitesimal response-loop torsion theorem

We can now state the first principal result.

Theorem 2.2 (Infinitesimal Response-Loop Torsion Theorem). *Let $x \in M$, let e^a be a C^2 nondegenerate solder form in a neighborhood of x , and let ω^a_b be a C^1 Lorentz connection. Let*

$$\gamma_\ell = \partial \Sigma_\ell \quad (78)$$

be a smooth family of contractible loops of linear size $O(\ell)$ shrinking to x . Let $P_a(s)$ be obtained by Lorentz parallel transport of a fixed initial covector

$$P_a = P_a(0). \quad (79)$$

Then

$$I_{\gamma_\ell}(P) = \oint_{\gamma_\ell} P_a(s) e^a = \int_{\Sigma_\ell} P_a T^a + O(\ell^3) \quad (80)$$

Equivalently,

$$I_{\gamma_\ell}(P) = \frac{1}{2} P_a T^a_{\mu\nu}(x) \Sigma_\ell^{\mu\nu} + O(\ell^3) \quad (81)$$

Consequently, for sufficiently small loops,

$$\log W_{\gamma_\ell}(P) = -\frac{i}{\hbar} \int_{\Sigma_\ell} P_a T^a + O(\ell^3) \quad (82)$$

Proof. Because the final result is Lorentz gauge invariant, we may perform the calculation in the local gauge

$$\omega^a_{b\mu}(x) = 0 \quad (83)$$

used in Lemma 2.1. Equation (77) gives

$$I_{\gamma_\ell}(P) = P_a \oint_{\gamma_\ell} e^a + O(\ell^3). \quad (84)$$

Ordinary Stokes' theorem, applied componentwise in this fixed local frame, yields

$$\oint_{\gamma_\ell} e^a = \int_{\Sigma_\ell} de^a. \quad (85)$$

Using the definition of torsion,

$$de^a = T^a - \omega^a{}_b \wedge e^b, \quad (86)$$

we obtain

$$P_a \oint_{\gamma_\ell} e^a = \int_{\Sigma_\ell} P_a T^a - \int_{\Sigma_\ell} P_a \omega^a{}_b \wedge e^b. \quad (87)$$

In the chosen gauge,

$$\omega^a{}_b = O(\ell) \quad (88)$$

on Σ_ℓ , whereas $e^a = O(1)$. Since the surface area is $O(\ell^2)$,

$$\int_{\Sigma_\ell} P_a \omega^a{}_b \wedge e^b = O(\ell^3). \quad (89)$$

Hence

$$I_{\gamma_\ell}(P) = \int_{\Sigma_\ell} P_a T^a + O(\ell^3), \quad (90)$$

which proves Eq. (80).

Expanding the smooth torsion two-form at x ,

$$T^a = \frac{1}{2} T^a{}_{\mu\nu}(x) dy^\mu \wedge dy^\nu + O(\ell), \quad (91)$$

gives

$$\int_{\Sigma_\ell} P_a T^a = \frac{1}{2} P_a T^a{}_{\mu\nu}(x) \Sigma_\ell^{\mu\nu} + O(\ell^3), \quad (92)$$

which proves Eq. (81).

Finally,

$$W_{\gamma_\ell}(P) = \exp \left[-\frac{i}{\hbar} I_{\gamma_\ell}(P) \right]. \quad (93)$$

Since

$$I_{\gamma_\ell}(P) = O(\ell^2), \quad (94)$$

the phase lies in a neighborhood of the identity for sufficiently small ℓ , so the logarithm may be taken on its continuous local branch. Substitution of Eq. (80) then gives Eq. (82). $\square$

2.6 Why curvature does not contaminate the leading torsion term

Theorem 2.2 does not assert that Lorentz curvature is absent from the closed-path comparison. Curvature enters the parallel transport of P and produces the usual infinitesimal Lorentz holonomy, schematically

$$\Delta_\gamma P \sim R \cdot P \Sigma = O(\ell^2). \quad (95)$$

What the theorem establishes is a statement about the order at which that holonomy contributes to the *translational response integral*.

The correction to the integrand induced by the curvature-generated change of P is

$$\delta P = O(\ell^2). \quad (96)$$

Because

$$e^a{}_\mu dx^\mu = O(\ell) \quad (97)$$

along a segment of the shrinking loop,

$$\oint_{\gamma_\ell} \delta P_a e^a = O(\ell^3). \quad (98)$$

Thus

torsion enters the response phase at $O(\ell^2)$, Lorentz-curvature contamination enters at $O(\ell^3)$ or higher

(99)

This order separation is local and concerns the particular scalar generator–displacement response considered here. It should not be read as a statement that curvature itself is a higher-order geometric effect. Curvature produces an $O(\ell^2)$ Lorentz holonomy of an internal vector. Its contribution becomes $O(\ell^3)$ only after that holonomy is inserted into the additional line integral defining the translational response.

2.7 Response-loop torsion criterion

The infinitesimal expansion becomes an operational characterization of the torsion-free sector once the generator family separates internal vectors.

Recall that the positive massive shell

$$\Sigma_{Q,x}^+ \subset V_x^* \quad (100)$$

spans the full internal dual space. Consequently,

$$P_a X^a = 0 \quad \text{for every } P \in \Sigma_{Q,x}^+ \quad (101)$$

implies

$$X^a = 0. \quad (102)$$

We can therefore state the following corollary.

Corollary 2.3 (Response-Loop Torsion Criterion). *Under the assumptions of Theorem 2.2, the following statements are equivalent at x :*

1.

$$T^a{}_x = 0; \quad (103)$$

2. for every fixed

$$P \in \Sigma_{Q,x}^+ \quad (104)$$

and every sufficiently small contractible loop $\gamma_\ell = \partial\Sigma_\ell$ based at x ,

$W_{\gamma_\ell}(P) = 1 + O(\ell^3)$

(105)

Equivalently,

$$\boxed{T^a_x = 0 \iff \text{all infinitesimal closed responses have no area-order phase}} \quad (106)$$

Proof. If

$$T^a_x = 0, \quad (107)$$

then Theorem 2.2 gives

$$I_{\gamma_\ell}(P) = O(\ell^3), \quad (108)$$

and hence

$$W_{\gamma_\ell}(P) = 1 + O(\ell^3). \quad (109)$$

Conversely, suppose Eq. (105) holds for every shell generator and every sufficiently small contractible loop. Since the loop shrinks to the identity, the possible $2\pi\hbar$ ambiguity of the exponential is absent: for sufficiently small ℓ ,

$$I_{\gamma_\ell}(P) \quad (110)$$

lies in a fixed neighborhood of zero. Therefore the vanishing of the area-order phase implies

$$P_a T^a_{\mu\nu}(x) \Sigma^{\mu\nu} = 0 \quad (111)$$

for every shell generator P and for every infinitesimal oriented surface bivector $\Sigma^{\mu\nu}$.

Shrinking loops may be chosen with arbitrary tangent area bivector at x . Hence

$$P_a T^a_{\mu\nu}(x) = 0 \quad (112)$$

for every pair μ, ν and every $P \in \Sigma_{Q,x}^+$. Because the massive shell spans V_x^* , Eq. (102) then gives

$$T^a_{\mu\nu}(x) = 0 \quad (113)$$

for all a, μ, ν . Thus

$$T^a_x = 0. \quad (114)$$

□

If the criterion holds at every point of an open region $\mathcal{U} \subset M$, then

$$T^a|_{\mathcal{U}} = 0. \quad (115)$$

2.8 Operational meaning of the criterion

The response-loop criterion changes the logical status of the torsion-free branch without changing its mathematical content.

In a standard first-order formulation one may impose, derive, or select

$$T^a = 0 \quad (116)$$

through a connection equation or through a restriction on the gravitational sector. In the present framework the equivalent local statement can instead be expressed in terms of the generator response:

$$\boxed{\text{torsion-free} \iff \text{absence of an area-order closed-response phase}} \quad (117)$$

This does not imply that the compact MS state

$$|\mathbf{D}|^2 + S^2 = 1 \quad (118)$$

forces torsion to vanish. Nor does local soldering alone select the torsion-free connection. Rather, once the operational response and Lorentz transport are present, torsion acquires a direct response interpretation.

The distinction is especially useful when the spinless and spinful sectors are compared later in this paper. In a structureless spinless branch one may impose the response-loop condition

$$W_{\gamma\ell}(P) = 1 + O(\ell^3) \quad (119)$$

and thereby select

$$T^a = 0. \quad (120)$$

When intrinsic spin is present, by contrast, the connection equation may generate

$$T^a \neq 0, \quad (121)$$

in which case the same loop observable records the induced torsional response rather than being required to vanish.

2.9 Relation to translational nonclosure

The geometrical interpretation of torsion as translational nonclosure is standard in Cartan geometry and gauge formulations of gravity. The present result should therefore not be interpreted as a new geometrical definition of torsion [?, ?].

What is specific to the present construction is the route by which the translation-valued nonclosure is converted into a scalar operational quantity. The solder form appearing in the torsion

$$T^a = De^a \quad (122)$$

is itself reconstructed from the generator–displacement response

$$\Theta(P, \xi) = P(e\xi), \quad (123)$$

and the same generator that resolves the local displacement response contracts the torsional nonclosure around a small loop:

$$\boxed{\Theta \longrightarrow e \longrightarrow T = De \longrightarrow P_a T^a \longrightarrow W_\gamma(P).} \quad (124)$$

Thus the new content is not

$$\text{“torsion measures nonclosure,”} \quad (125)$$

which is classical, but rather

$$\boxed{\text{the same operational generator that reconstructs soldering also resolves its infinitesimal translational nonclosure}} \quad (126)$$

This observation will be used in two different ways below. First, in the spinless sector it supplies an operational criterion for selecting the torsion-free branch. Second, when spin is introduced, the nonzero torsion generated by the spin current becomes directly visible through the closed-response phase.

Before introducing gravitational constitutive dynamics, however, we next extract another purely kinematical consequence of the same local response: the operational construction of simple internal area bivectors.

3 Operational Area Bivectors and Simplicity

The preceding section showed that the local generator–displacement response determines the solder map

$$e_x : T_x M \longrightarrow V_x \quad (127)$$

through

$$\Theta_x(P, \xi) = P(e_x \xi). \quad (128)$$

We now show that the same response determines an intrinsic area geometry on the internal Lorentz space.

The resulting bivectors are simple by construction. This statement should be distinguished from the role of simplicity constraints in BF and Plebanski formulations. Here the bivector is not introduced as an independent variable and subsequently constrained to be of the form $e \wedge e$. Rather, the solder map has already been reconstructed from the operational response, and the geometric bivector is then obtained as a composite object.

3.1 Operational edge vectors

Let

$$\xi \in T_x M \quad (129)$$

be an infinitesimal spacetime displacement. We associate with ξ the internal Lorentz vector

$$E_\xi := e_x(\xi) \in V_x \quad (130)$$

In components,

$$E_\xi^a = e^a{}_\mu \xi^\mu. \quad (131)$$

Equation (128) then becomes

$$\Theta_x(P, \xi) = P_a E_\xi^a \quad (132)$$

Thus E_ξ is not an independently assumed internal edge variable. Its pairing with every physically available massive generator is already specified by the operational response.

Because the positive massive shell

$$\Sigma_{Q,x}^+ \subset V_x^* \quad (133)$$

spans the internal dual space, the values

$$P \longmapsto \Theta_x(P, \xi) \quad (134)$$

determine E_ξ uniquely.

Proposition 3.1 (Operational reconstruction of an internal edge). *Assume that the massive generator shell $\Sigma_{Q,x}^+$ spans V_x^* , and that the local response satisfies*

$$\Theta_x(P, \xi) = P(e_x \xi). \quad (135)$$

Then, for every $\xi \in T_x M$, there exists a unique $E_\xi \in V_x$ such that

$$\Theta_x(P, \xi) = P(E_\xi) \quad (136)$$

for every $P \in \Sigma_{Q,x}^+$. This vector is

$$E_\xi = e_x(\xi) \quad (137)$$

Proof. Existence follows from the reconstructed solder map by taking

$$E_\xi = e_x(\xi). \quad (138)$$

For uniqueness, suppose that E_ξ and E'_ξ give the same response for every shell generator. Then

$$P(E_\xi - E'_\xi) = 0 \quad (139)$$

for every $P \in \Sigma_{Q,x}^+$. Since the shell spans V_x^* , the same equation holds for every covector in V_x^* . Hence

$$E_\xi - E'_\xi = 0. \quad (140)$$

□

Linearity of the solder map immediately gives

$$E_{\lambda\xi + \mu\zeta} = \lambda E_\xi + \mu E_\zeta, \quad (141)$$

so the operational edge construction preserves the linear infinitesimal displacement structure.

3.2 Operational area bivectors

Let

$$\xi, \zeta \in T_x M. \quad (142)$$

We define their oriented internal area bivector by

$$\boxed{\mathbb{B}_{\xi\zeta} := E_\xi \wedge E_\zeta \in \Lambda^2 V_x} \quad (143)$$

In components we use the convention

$$\boxed{\mathbb{B}_{\xi\zeta}^{ab} = E_\xi^a E_\zeta^b - E_\zeta^a E_\xi^b = 2E_\xi^{[a} E_\zeta^{b]}} \quad (144)$$

The corresponding bivector-valued spacetime two-form is

$$\boxed{B^{ab} = e^a \wedge e^b} \quad (145)$$

Its evaluation on the oriented infinitesimal two-surface generated by ξ and ζ gives

$$\begin{aligned} B^{ab}(\xi, \zeta) &= e^a(\xi)e^b(\zeta) - e^a(\zeta)e^b(\xi) \\ &= \mathbb{B}_{\xi\zeta}^{ab}. \end{aligned} \quad (146)$$

Thus the local two-form B^{ab} packages all operational infinitesimal area bivectors.

The map

$$(\xi, \zeta) \mapsto \mathbb{B}_{\xi\zeta} \quad (147)$$

is bilinear and antisymmetric:

$$\mathbb{B}_{\xi\zeta} = -\mathbb{B}_{\zeta\xi}, \quad (148)$$

$$\mathbb{B}_{\lambda\xi + \mu\chi, \zeta} = \lambda\mathbb{B}_{\xi\zeta} + \mu\mathbb{B}_{\chi\zeta}. \quad (149)$$

These properties follow directly from the linearity of the operational edge map.

3.3 Simplicity

A bivector

$$X \in \Lambda^2 V_x \quad (150)$$

is called simple or decomposable if there exist vectors $U, V \in V_x$ such that

$$X = U \wedge V. \quad (151)$$

In four dimensions this is equivalent to

$$X \wedge X = 0, \quad (152)$$

or, in internal components,

$$X^{[ab} X^{cd]} = 0. \quad (153)$$

For the operational area bivector, simplicity is immediate:

$$\begin{aligned} \mathbb{B}_{\xi\zeta} \wedge \mathbb{B}_{\xi\zeta} &= (E_\xi \wedge E_\zeta) \wedge (E_\xi \wedge E_\zeta) \\ &= 0. \end{aligned} \quad (154)$$

Equivalently,

$$\boxed{\mathbb{B}_{\xi\zeta}^{[ab} \mathbb{B}_{\xi\zeta}^{cd]} = 0} \quad (155)$$

Proposition 3.2 (Operational simplicity). *Let the local generator–displacement response satisfy the operational soldering assumptions, and let*

$$E_\xi = e_x(\xi) \quad (156)$$

be the corresponding reconstructed internal edge vector. Then every infinitesimal operational area bivector

$$\mathbb{B}_{\xi\zeta} = E_\xi \wedge E_\zeta \quad (157)$$

is simple. In four internal dimensions it therefore satisfies

$$\boxed{\mathbb{B}_{\xi\zeta}^{[ab} \mathbb{B}_{\xi\zeta}^{cd]} = 0} \quad (158)$$

Thus simplicity is inherited from the operational soldering map rather than imposed as an independent constraint on a primitive bivector variable.

Proof. By definition,

$$\mathbb{B}_{\xi\zeta} = E_\xi \wedge E_\zeta, \quad (159)$$

so it is decomposable. Hence

$$\mathbb{B}_{\xi\zeta} \wedge \mathbb{B}_{\xi\zeta} = E_\xi \wedge E_\zeta \wedge E_\xi \wedge E_\zeta = 0 \quad (160)$$

by antisymmetry of the exterior product. In four dimensions this is equivalent to

$$\mathbb{B}_{\xi\zeta}^{[ab} \mathbb{B}_{\xi\zeta}^{cd]} = 0. \quad (161)$$

□

3.4 Cross-simplicity from a shared operational edge

The same construction gives the elementary cross-simplicity property. Suppose two oriented infinitesimal areas share the displacement ξ :

$$\mathbb{B}_1 = E_\xi \wedge E_\zeta, \quad \mathbb{B}_2 = E_\xi \wedge E_\chi. \quad (162)$$

Then

$$\begin{aligned} \mathbb{B}_1 \wedge \mathbb{B}_2 &= E_\xi \wedge E_\zeta \wedge E_\xi \wedge E_\chi \\ &= 0. \end{aligned} \quad (163)$$

In components,

$$\boxed{\mathbb{B}_1^{[ab} \mathbb{B}_2^{cd]} = 0} \quad (164)$$

Corollary 3.3 (Operational cross-simplicity). *If two operational area bivectors share a reconstructed internal edge,*

$$\mathbb{B}_1 = E_\xi \wedge E_\zeta, \quad \mathbb{B}_2 = E_\xi \wedge E_\chi, \quad (165)$$

then

$$\boxed{\mathbb{B}_1 \wedge \mathbb{B}_2 = 0} \quad (166)$$

Equivalently,

$$\boxed{\mathbb{B}_1^{[ab} \mathbb{B}_2^{cd]} = 0} \quad (167)$$

The result is purely kinematical. It requires no field equation, curvature condition, torsion condition, or constitutive law.

3.5 Direct reconstruction from response derivatives

Although the bivector construction is most naturally written through the already reconstructed solder map, it can also be expressed directly in terms of the generator dependence of the response.

Choose an internal dual basis

$$\{\varepsilon^a\} \subset V_x^* \quad (168)$$

and let

$$P(s) = P_0 + s_a \varepsilon^a \quad (169)$$

be a local affine extension in generator space. Since the response is linear in P ,

$$\frac{\partial}{\partial s_a} \Theta_x(P(s), \xi) = \varepsilon^a(E_\xi). \quad (170)$$

In the dual basis this gives

$$\boxed{E_\xi^a = \frac{\partial \Theta_x(P, \xi)}{\partial P_a}} \quad (171)$$

Consequently,

$$\boxed{\mathbb{B}_{\xi\zeta}^{ab} = \frac{\partial \Theta(P, \xi)}{\partial P_a} \frac{\partial \Theta(P, \zeta)}{\partial P_b} - \frac{\partial \Theta(P, \xi)}{\partial P_b} \frac{\partial \Theta(P, \zeta)}{\partial P_a}} \quad (172)$$

Equation (172) is useful for the operational interpretation, but it should not be mistaken for an additional dynamical relation. It is simply the bivector $e(\xi) \wedge e(\zeta)$ expressed through the generator derivative of the response.

If one works directly with the local phase

$$W(P, \xi) = \exp \left[-\frac{i}{\hbar} \Theta(P, \xi) \right], \quad (173)$$

then at the identity in generator space,

$$i\hbar \frac{\partial W(P, \xi)}{\partial P_a} \Big|_{P=0} = E_\xi^a. \quad (174)$$

Hence the antisymmetrized product of two first response derivatives recovers the same area bivector:

$$\boxed{-\hbar^2 \left[\frac{\partial W(P_1, \xi)}{\partial P_{1a}} \frac{\partial W(P_2, \zeta)}{\partial P_{2b}} - (a \leftrightarrow b) \right]_{P_1=P_2=0}} = \mathbb{B}_{\xi\zeta}^{ab} \quad (175)$$

The use of $P = 0$ in Eq. (174) refers to the linear extension of the response away from the physical massive shell. No physical massive state at $P = 0$ is being assumed.

3.6 Relation to Plebanski simplicity

The algebraic statements

$$B = e \wedge e, \quad B \wedge B = 0 \quad (176)$$

for individual simple bivectors are familiar from tetrad and Plebanski formulations of gravity. In constrained BF theories, one begins with an independent Lie-algebra-valued two-form B and imposes simplicity constraints in order to restrict the theory to a geometric sector.

The logical direction adopted here is different:

$$\boxed{\begin{aligned} \text{generator-displacement response} &\implies e \\ &\implies E_\xi = e(\xi) \\ &\implies \mathbb{B}_{\xi\zeta} = E_\xi \wedge E_\zeta \\ &\implies \text{simplicity and cross-simplicity.} \end{aligned}} \quad (177)$$

Thus no independent simplicity condition is required at the kinematical level considered here.

This distinction is important for the scope of the present work. Operational simplicity does not constitute a new solution of the Plebanski simplicity constraints, nor does it determine a BF quantization or a spin-foam amplitude. It states only that once nondegenerate soldering has been reconstructed from the operational response, the infinitesimal area variables derived from that response necessarily lie in the geometric simple-bivector sector.

3.7 What the area construction does not determine

The construction of

$$B^{ab} = e^a \wedge e^b \quad (178)$$

uses only the local operational geometry. It does not determine:

1. the Lorentz connection ω^{ab} beyond whatever transport structure has separately been supplied;
2. the curvature

$$R^{ab} = d\omega^{ab} + \omega^a_c \wedge \omega^{cb}; \quad (179)$$

3. the gravitational excitation conjugate to curvature;
4. the normalization of the gravitational action;
5. the relative weighting of parity-even and parity-odd first-order curvature couplings.

In particular, knowledge of the operational bivector $B = e \wedge e$ does not determine a map

$$\mathfrak{G} : \Lambda^2 V \longrightarrow \Lambda^2 V \quad (180)$$

relating geometric area to gravitational excitation.

This yields the central separation required for the next section:

$$\boxed{\underbrace{\Theta \implies e \implies B = e \wedge e}_{\text{operational kinematics}} \not\Rightarrow \underbrace{H = \mathfrak{G}(B)}_{\text{gravitational constitutive law}}} \quad (181)$$

The operational response therefore fixes the geometric bivector on which a first-order gravitational constitutive law may act, but it does not fix that law itself. The classification of the remaining Lorentz-equivariant constitutive freedom is the subject of the next section.

4 Lorentz-Equivariant Constitutive Rigidity

The preceding sections reconstruct two pieces of first-order geometric information directly from the operational response. First, $\Theta(P, \xi)$ determines the solder form e^a , and hence the Lorentzian metric

$$g = e^* \eta. \quad (182)$$

Second, two infinitesimal displacements determine the simple internal area bivector

$$B^{ab} = e^a \wedge e^b. \quad (183)$$

Neither statement, however, specifies how this kinematical area variable is to couple to Lorentz curvature. The purpose of the present section is to classify the remaining constitutive freedom under a minimal set of local assumptions.

Throughout this section, V denotes an oriented real four-dimensional Lorentz vector space with signature

$$\eta = \text{diag}(-1, +1, +1, +1), \quad (184)$$

and

$$G = SO^+(1, 3) \quad (185)$$

acts on V in the defining representation and on $\Lambda^2 V$ by the induced bivector representation.

The Hodge operator used below acts on internal Lorentz bivector indices only. With the orientation fixed by $\epsilon_{0123} = +1$, we define

$$(\star X)^{ab} := \frac{1}{2} \epsilon^{ab}_{cd} X^{cd}, \quad X \in \Lambda^2 V. \quad (186)$$

For Lorentz signature $(- + + +)$,

$$\boxed{\star^2 = -I} \quad (187)$$

on $\Lambda^2 V$. This internal Hodge operator should not be confused with the spacetime Hodge operator acting on differential-form indices.

4.1 The constitutive problem

Let

$$\mathfrak{C}_x : \Lambda^2 V_x \longrightarrow \Lambda^2 V_x \quad (188)$$

be a real linear map at a spacetime point x . We interpret $\mathfrak{C}_x$ as a local constitutive map relating the operational area bivector B to a curvature excitation H :

$$H_{ab} = \mathfrak{C}(B)_{ab} \quad (189)$$

Here both B^{ab} and H^{ab} are spacetime two-forms carrying an antisymmetric pair of internal Lorentz indices, while $\mathfrak{C}$ acts only on the internal bivector indices.

The constitutive map is required to be Lorentz equivariant:

$$\mathfrak{C}(\Lambda \cdot X) = \Lambda \cdot \mathfrak{C}(X) \quad \forall \Lambda \in SO^+(1, 3), \quad X \in \Lambda^2 V. \quad (190)$$

Equivalently,

$$\mathfrak{C} \in \text{End}_{SO^+(1,3)}(\Lambda^2 V), \quad (191)$$

the commutant of the proper orthochronous Lorentz action on real bivectors.

The following lemma determines this commutant completely.

Lemma 4.1 (Lorentz bivector commutant). *Let V be an oriented real four-dimensional Lorentz vector space. Then*

$$\text{End}_{SO^+(1,3)}(\Lambda^2 V) = \text{span}_{\mathbb{R}}\{I, \star\} \quad (192)$$

In particular, every real Lorentz-equivariant endomorphism of $\Lambda^2 V$ has the unique form

$$\mathfrak{C} = \alpha I + \beta \star, \quad \alpha, \beta \in \mathbb{R}. \quad (193)$$

Proof. Complexify the real bivector representation:

$$\Lambda^2 V_{\mathbb{C}} := \Lambda^2 V \otimes_{\mathbb{R}} \mathbb{C}. \quad (194)$$

Since $\star^2 = -I$, the complexified Hodge operator has eigenvalues $\pm i$, and therefore

$$\Lambda^2 V_{\mathbb{C}} = \Lambda_+^2 \oplus \Lambda_-^2, \quad (195)$$

where

$$\Lambda_{\pm}^2 := \{X \in \Lambda^2 V_{\mathbb{C}} : \star X = \pm iX\}. \quad (196)$$

Each sector has complex dimension three. Under the connected Lorentz group, the two sectors furnish the two inequivalent irreducible complex bivector representations conventionally denoted by $(1, 0)$ and $(0, 1)$.

Let

$$\mathfrak{C}_{\mathbb{C}} : \Lambda^2 V_{\mathbb{C}} \longrightarrow \Lambda^2 V_{\mathbb{C}} \quad (197)$$

be the complexification of a real Lorentz-equivariant map $\mathfrak{C}$. Because $\mathfrak{C}_{\mathbb{C}}$ commutes with the Lorentz action, Schur's lemma implies that it acts by scalar multiplication on each irreducible sector:

$$\mathfrak{C}_{\mathbb{C}}|_{\Lambda_+^2} = c_+ I, \quad \mathfrak{C}_{\mathbb{C}}|_{\Lambda_-^2} = c_- I, \quad (198)$$

for some $c_{\pm} \in \mathbb{C}$.

Complex conjugation interchanges Λ_+^2 and Λ_-^2 . Since the original map $\mathfrak{C}$ is real, $\mathfrak{C}_{\mathbb{C}}$ commutes with this conjugation, and hence

$$c_- = \overline{c_+}. \quad (199)$$

Write

$$c_+ = \alpha + i\beta, \quad c_- = \alpha - i\beta, \quad \alpha, \beta \in \mathbb{R}. \quad (200)$$

On Λ_+^2 , the operator $\alpha I + \beta \star$ has eigenvalue $\alpha + i\beta$, whereas on Λ_-^2 it has eigenvalue $\alpha - i\beta$. Therefore

$$\mathfrak{C}_{\mathbb{C}} = \alpha I + \beta \star. \quad (201)$$

Restricting back to the real bivector space gives Eq. (193). Uniqueness follows from the linear independence of I and $\star$ over $\mathbb{R}$. $\square$

The preceding lemma is a standard representation-theoretic fact about four-dimensional Lorentz bivectors. Its role in the present construction is not to introduce a new classification of first-order gravity. Rather, it identifies the complete local linear constitutive freedom that remains *after* the operational response has already reconstructed the simple geometric area variable $B = e \wedge e$.

We may state this consequence separately.

Theorem 4.2 (Lorentz-equivariant constitutive rigidity). *Let e be a nondegenerate operational solder form on an oriented four-dimensional Lorentzian geometry, and let*

$$B^{ab} = e^a \wedge e^b. \quad (202)$$

Suppose the curvature excitation is related to B by a local, real, linear constitutive law

$$H = \mathfrak{C}(B) \quad (203)$$

which is equivariant under $SO^+(1, 3)$. Then pointwise

$$\boxed{H = \alpha B + \beta \star B} \quad (204)$$

for real scalar coefficients α, β .

If no additional spacetime-dependent constitutive scalar fields or external scalar backgrounds are admitted, the coefficients are fixed universal constants within the constitutive sector under consideration.

Proof. The pointwise result follows immediately from Lemma 4.1. Lorentz equivariance alone allows the coefficients to be Lorentz scalars $\alpha(x)$ and $\beta(x)$. If they are permitted to vary independently over spacetime, they constitute additional scalar constitutive data. The restricted sector considered here contains no such additional fields or backgrounds, and therefore treats α and β as fixed constitutive parameters. $\square$

4.2 Invertibility and constitutive parameters

The Lorentzian identity $\star^2 = -I$ immediately gives

$$(\alpha I + \beta \star)(\alpha I - \beta \star) = (\alpha^2 + \beta^2)I. \quad (205)$$

Hence every nonzero real constitutive map in the family is invertible.

Corollary 4.3 (Nondegeneracy of the real constitutive family). *If*

$$(\alpha, \beta) \neq (0, 0), \quad (206)$$

then

$$\mathfrak{C}^{-1} = \frac{\alpha I - \beta \star}{\alpha^2 + \beta^2} \quad (207)$$

It is useful to separate the two kinds of constitutive information contained in (α, β) . The positive quantity

$$\sqrt{\alpha^2 + \beta^2} \quad (208)$$

sets an overall constitutive magnitude, whereas the projective ratio

$$[\alpha : \beta] \quad (209)$$

specifies a direction in the two-dimensional algebra $\text{span}_{\mathbb{R}}\{I, \star\}$.

When $\beta \neq 0$, one may introduce an Immirzi-type ratio by

$$\frac{1}{\gamma} := \frac{\alpha}{\beta} \quad (210)$$

Then

$$\mathfrak{C} = \beta \left(\star + \frac{1}{\gamma} I \right) \quad (211)$$

If the overall normalization is conventionally written as

$$\beta = \frac{1}{2\kappa}, \quad (212)$$

this becomes

$$\mathfrak{C} = \frac{1}{2\kappa} \left(\star + \frac{1}{\gamma} I \right) \quad (213)$$

up to the overall normalization convention adopted for the first-order action.

At the present stage, neither κ nor γ has been determined. Equation (213) is only a convenient reparameterization of the two independent real constitutive coefficients.

4.3 First-order curvature coupling

Let

$$R^{ab} = d\omega^{ab} + \omega^a_c \wedge \omega^{cb} \quad (214)$$

be the curvature of a metric-compatible Lorentz connection. Within the restricted local sector that is linear in curvature and contains no additional gravitational fields, the constitutive excitation enters through

$$S_{\mathfrak{C}}[e, \omega] = \frac{1}{2} \int_M H_{ab} \wedge R^{ab} + S_{\Lambda}[e] \quad (215)$$

The volume contribution may be written as

$$S_\Lambda[e] = -\lambda_\Lambda \int_M \text{vol}_e, \quad (216)$$

where

$$\text{vol}_e = \frac{1}{4!} \epsilon_{abcd} e^a \wedge e^b \wedge e^c \wedge e^d. \quad (217)$$

Substituting $\mathfrak{C} = \alpha I + \beta \star$ gives

$$S_\mathfrak{C} = \frac{\alpha}{2} \int_M B_{ab} \wedge R^{ab} + \frac{\beta}{2} \int_M (\star B)_{ab} \wedge R^{ab} + S_\Lambda. \quad (218)$$

Using the invariance of the internal bivector pairing under the Hodge operator,

$$(\star B)_{ab} R^{ab} = B_{ab} (\star R)^{ab}, \quad (219)$$

one may equivalently write

$$S_\mathfrak{C} = \frac{\alpha}{2} \int_M B_{ab} \wedge R^{ab} + \frac{\beta}{2} \int_M B_{ab} \wedge (\star R)^{ab} + S_\Lambda \quad (220)$$

Since

$$B^{ab} = e^a \wedge e^b, \quad (221)$$

the first term becomes

$$\int_M e_a \wedge e_b \wedge R^{ab}, \quad (222)$$

while the second is

$$\frac{1}{2} \int_M \epsilon_{abcd} e^a \wedge e^b \wedge R^{cd}. \quad (223)$$

These are respectively the identity/Holst and Hodge-dual/Palatini–Einstein–Cartan directions, up to conventional overall coefficients and sign choices.

Thus the operational and constitutive stages of the construction are sharply separated:

$$\boxed{\begin{array}{l} \Theta \implies e \implies B = e \wedge e, \\ \text{local real linear Lorentz covariance} \implies \mathfrak{C} = \alpha I + \beta \star. \end{array}} \quad (224)$$

The first line fixes the geometric area variable. The second line classifies the allowed constitutive map but leaves its two coefficients undetermined.

This distinction is essential. The result does not state that operational soldering derives the Einstein–Cartan action, nor that Lorentz covariance determines Newton’s constant or the Immirzi parameter. It states instead that, within the local real linear first-order sector considered here, all remaining Lorentz-equivariant area–curvature response is confined to the two-dimensional algebra generated by I and $\star$.

The standard Einstein–Cartan and Holst structures therefore appear here as the two basis directions of the residual constitutive freedom after the operational response has already determined the simple bivector B . The next section asks which parts of this freedom are visible in the spinless torsion-free bulk theory and which remain operationally underdetermined.

5 Spinless Sector and Constitutive Underdetermination

The preceding sections separate two structures that should not be identified. The operational response reconstructs the solder form and therefore the simple bivector

$$B^{ab} = e^a \wedge e^b, \quad (225)$$

whereas local Lorentz covariance restricts, but does not uniquely fix, the linear constitutive map relating this area variable to curvature excitation:

$$H_{ab} = \mathfrak{C}(B)_{ab}, \quad \mathfrak{C} = \alpha I + \beta \star. \quad (226)$$

Here the internal Hodge operator satisfies

$$\star^2 = -I \quad (227)$$

on real Lorentzian bivectors.

The purpose of the present section is to determine how much of the constitutive freedom in Eq. (226) remains visible in the structureless spinless sector. The result has two parts. First, on the response-loop torsion-free branch, the identity component of the constitutive map is invisible in the classical bulk equations. Second, neither the relative identity–Hodge-dual coefficient nor the overall gravitational response scale is determined by the operational data reconstructed in the preceding sections.

5.1 The torsion-free response sector

We restrict first to configurations satisfying the infinitesimal response-loop criterion of Sec. 2. Thus, for every sufficiently small contractible loop $\gamma = \partial\Sigma$ and every generator P in a shell spanning the internal dual space,

$$W_\gamma(P) = 1 + O(\ell^3). \quad (228)$$

By the response-loop torsion theorem, this condition is equivalent, under the stated local assumptions, to

$$\boxed{T^a = 0} \quad (229)$$

For a nondegenerate solder form and a metric-compatible Lorentz connection, Eq. (229) selects the unique Levi–Civita spin connection

$$\omega^{ab} = \omega_{\text{LC}}^{ab}[e]. \quad (230)$$

The curvature is then

$$R^{ab} = R^{ab}[\omega_{\text{LC}}(e)]. \quad (231)$$

It is important that Eq. (229) is not being derived here from the compact MS state alone. It is the geometrical content of the operational closed-response condition (228). The present section asks what constitutive information remains once that operational sector has been selected.

5.2 Nieh–Yan identity and the Holst direction

The relevant identity is the Nieh–Yan relation

$$d(e_a \wedge T^a) = T_a \wedge T^a - e_a \wedge e_b \wedge R^{ab}. \quad (232)$$

On the torsion-free branch,

$$T^a = 0, \quad (233)$$

both the left-hand side and the first term on the right-hand side of Eq. (232) vanish. Hence

$$\boxed{e_a \wedge e_b \wedge R^{ab} = 0} \quad (234)$$

Equivalently,

$$\boxed{B_{ab} \wedge R^{ab} = 0} \quad (235)$$

The same statement follows from the algebraic first Bianchi identity for the Levi–Civita connection. Equation (235) shows that the identity component of the constitutive map,

$$\mathfrak{C}_{\text{id}} = \alpha I, \quad (236)$$

produces no independent torsion-free bulk density.

By contrast, the Hodge-dual component gives

$$B^{ab} \wedge (\star R)_{ab} = \frac{1}{2} \epsilon_{ab}{}^{cd} e^a \wedge e^b \wedge R_{cd}, \quad (237)$$

which is the usual Palatini–Einstein–Cartan curvature density.

Thus, for

$$S_{\text{g}}[e, \omega] = \frac{1}{2} \int B^{ab} \wedge (\alpha R_{ab} + \beta (\star R)_{ab}) - \Lambda \int \text{vol}_e, \quad (238)$$

the torsion-free reduction yields

$$S_{\text{g}}[e, \omega_{\text{LC}}(e)] = \frac{\beta}{2} \int B^{ab} \wedge (\star R)_{ab} - \Lambda \int \text{vol}_e, \quad (239)$$

up to boundary and topological terms associated with the chosen first-order completion.

Equation (239) should be interpreted at the level of the torsion-free branch. The identity/Holst term is not being claimed to be absent from the full first-order variational problem. Rather, once the nondegenerate connection equation has selected the torsion-free branch, its classical bulk contribution vanishes and the reduced field equations coincide with those of the Einstein–Cartan/Palatini sector.

5.3 Spinless Einstein equivalence class

We can now state the relevant equivalence result.

Theorem 5.1 (Spinless Einstein equivalence class). *Let M be an oriented four-dimensional spacetime equipped with a nondegenerate operational solder form e^a , and let ω^{ab} be a metric-compatible Lorentz connection. Assume:*

1. *the operational area bivector is*

$$B^{ab} = e^a \wedge e^b; \quad (240)$$

2. the gravitational constitutive relation is local, real, linear, and Lorentz equivariant,

$$H = (\alpha I + \beta \star)B; \quad (241)$$

3. no additional gravitational fields enter the local first-order bulk action;

4. the curvature dependence is at most linear;

5. the response-loop torsion-free condition holds,

$$T^a = 0; \quad (242)$$

6.

$$\beta \neq 0. \quad (243)$$

Then the classical torsion-free bulk equations obtained from the family

$$\mathfrak{C}_{\alpha,\beta} = \alpha I + \beta \star \quad (244)$$

are equivalent, after an overall normalization of the Hodge-dual component, to the Einstein equations with cosmological constant. The parameter α does not modify the torsion-free spinless bulk equations.

Proof. On the response-loop torsion-free branch,

$$T^a = 0. \quad (245)$$

Equation (232) then gives

$$B_{ab} \wedge R^{ab} = 0. \quad (246)$$

Hence the identity component of $\mathfrak{C}_{\alpha,\beta}$ contributes no independent torsion-free bulk density.

The remaining curvature term is

$$\frac{\beta}{2} B^{ab} \wedge (\star R)_{ab}, \quad (247)$$

which is the standard Palatini–Einstein–Cartan term multiplied by the nonzero constant β . Variation of the reduced action with respect to the coframe therefore gives the Einstein equation with an overall coupling proportional to β^{-1} , together with the cosmological contribution if the volume term is present. The parameter α drops out of the torsion-free spinless bulk equations. Thus all members of the family (244) with fixed nonzero Einstein component belong to the same classical spinless Einstein equivalence class. $\square$

The theorem is deliberately restricted. It does not state that the Holst coefficient is physically irrelevant in every sector. In particular, once intrinsic spin sources torsion, or once independent connection dynamics, boundaries, or quantum effects are retained, the identity component of the constitutive map can become physically distinguishable.

It is useful to parameterize the same family as

$$\boxed{\mathfrak{C} = \frac{1}{2\kappa} \left(\star + \frac{1}{\gamma} I \right)} \quad (248)$$

whenever the Hodge-dual component is nonzero. In this convention, κ controls the overall gravitational response scale and γ parametrizes the relative identity/Hodge-dual orientation. The precise correspondence with (α, β) is

$$\boxed{\beta = \frac{1}{2\kappa}, \quad \alpha = \frac{1}{2\kappa\gamma}} \quad (249)$$

Within the torsion-free spinless bulk sector, Eq. (248) therefore contains two logically different kinds of constitutive freedom:

$\kappa :$ overall gravitational response magnitude, $\gamma :$ relative identity–Hodge constitutive orientation.	(250)
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The first enters the Einstein equations through their overall normalization. The second is invisible in the classical torsion-free bulk equations.

5.4 Constitutive underdetermination

We now formulate the stronger logical statement relevant to the MS reconstruction.

Let

$$\mathfrak{A}_{\text{op}} \tag{251}$$

denote the collection of operational structures fixed before a constitutive gravitational law is chosen. For the present purpose, this includes

$$\mathfrak{A}_{\text{op}} = \{\text{compact MS state structure, } \Theta, e, g, B = e \wedge e, W_{\text{local}}, \text{Lorentz transport, response-loop torsion criterion}\}. \tag{252}$$

The precise list may be shortened or enlarged without affecting the argument below, provided none of its elements already specifies the curvature–excitation normalization.

Theorem 5.2 (Constitutive underdetermination). *The operational data $\mathfrak{A}_{\text{op}}$ do not uniquely determine either the gravitational coupling κ or the relative Holst/Immirzi parameter γ . More precisely, there exist continuous families of distinct first-order constitutive theories*

$$\mathfrak{C}_{\kappa, \gamma} = \frac{1}{2\kappa} \left(\star + \frac{1}{\gamma} I \right) \tag{253}$$

with different values of (κ, γ) that share the same

$$\text{MS state structure, } \Theta, \quad e, \quad g, \quad B, \quad W_{\text{local}}, \tag{254}$$

and, when restricted to the response-loop torsion-free spinless sector, the same Einstein tensor structure.

Proof. Fix a nondegenerate solder form e , the associated metric $g = e^* \eta$, the local response

$$\Theta(P, \xi) = P(e\xi), \tag{255}$$

and hence the operational bivector

$$B = e \wedge e. \tag{256}$$

None of these definitions contains either κ or γ .

Now choose two pairs of constitutive parameters

$$(\kappa_1, \gamma_1) \neq (\kappa_2, \gamma_2) \tag{257}$$

and define

$$\mathfrak{C}_1 = \frac{1}{2\kappa_1} \left(\star + \frac{1}{\gamma_1} I \right), \quad \mathfrak{C}_2 = \frac{1}{2\kappa_2} \left(\star + \frac{1}{\gamma_2} I \right). \tag{258}$$

Both constitutive maps act on exactly the same operationally reconstructed bivector B . They therefore define different curvature excitations,

$$H^{(1)} = \mathfrak{C}_1(B), \quad H^{(2)} = \mathfrak{C}_2(B), \quad (259)$$

without altering the underlying Θ , e , g , or local response algebra.

In the torsion-free spinless branch,

$$B_{ab} \wedge R^{ab} = 0, \quad (260)$$

so the difference in γ does not alter the classical bulk Einstein tensor structure. A change in κ , however, changes the normalization relating the same geometrical curvature to a fixed physical matter source. Therefore different (κ, γ) represent genuinely different constitutive completions of the same operational kinematics.

It follows that the operational structures listed in Eq. (252) cannot uniquely determine either parameter. $\square$

The theorem may be summarized by the two non-implications

$$\boxed{\mathfrak{A}_{\text{op}} \not\Rightarrow G} \quad (261)$$

and

$$\boxed{\mathfrak{A}_{\text{op}} \not\Rightarrow \gamma} \quad (262)$$

The first statement remains true even if the response calibration has already fixed the absolute space-time metric scale. Metric calibration and gravitational constitutive normalization are logically different operations.

Indeed, after the Einstein normalization

$$\kappa = \frac{8\pi G}{c^4}, \quad (263)$$

the Hodge-dual constitutive coefficient is

$$\beta = \frac{c^4}{16\pi G}. \quad (264)$$

Thus

$$\boxed{\frac{1}{\kappa} \sim \frac{c^4}{G}} \quad (265)$$

measures the magnitude of the curvature excitation associated with a given operational area bivector.

This motivates the constitutive interpretation

$$\boxed{G^{-1} \text{ is a gravitational response stiffness}} \quad (266)$$

up to the conventional factors of c and 8π . The statement is interpretive rather than predictive: it identifies the role played by G within the reconstructed hierarchy, but it does not determine its numerical magnitude.

5.5 Why the particle scale cannot fix the gravitational coupling

The underdetermination theorem also excludes a tempting identification between the intrinsic scale of an individual massive MS state and the universal gravitational coupling.

For a free massive particle, the compact-state reconstruction contains the intrinsic generator scale

$$Q_{\text{particle}} = mc. \quad (267)$$

This parameter is species dependent. By contrast,

$$G \quad (268)$$

is a universal constitutive coupling of the gravitational sector. Nothing in

$$\Theta(P, \xi) = P(e\xi) \quad (269)$$

or in the operational loop criterion relates the particle-dependent radial shell scale to the normalization of $\mathfrak{C}$.

Consequently,

$$\boxed{Q_{\text{particle}} \not\Rightarrow \kappa} \quad (270)$$

within the present framework. Likewise,

$$\boxed{(\hbar, c, Q_{\text{particle}}) \not\Rightarrow G} \quad (271)$$

without an additional microscopic principle.

The appearance of

$$\hbar \quad (272)$$

in the response phase

$$W_\gamma(P) = \exp \left[-\frac{i}{\hbar} \oint_\gamma P_a e^a \right] \quad (273)$$

does not change this conclusion. It fixes the conversion between action response and phase; it does not determine the independent constitutive slope relating operational area geometry to curvature excitation.

5.6 The role of the cosmological term

The same logical separation applies to the cosmological constant. The operational soldering determines the oriented volume form

$$\text{vol}_e = \frac{1}{4!} \epsilon_{abcd} e^a \wedge e^b \wedge e^c \wedge e^d, \quad (274)$$

but it does not determine the coefficient multiplying this volume in the gravitational action.

Thus the local first-order action may contain

$$S_\Lambda = -\frac{\Lambda}{\kappa} \int \text{vol}_e \quad (275)$$

without changing any of the preceding operational reconstruction of e , g , or B .

Accordingly,

$$\boxed{\mathfrak{A}_{\text{op}} \not\Rightarrow \Lambda} \quad (276)$$

The cosmological parameter therefore remains an independent constitutive/dynamical datum of the low-derivative gravitational completion.

5.7 Logical endpoint of the spinless sector

The results of this section establish a sharp boundary between operationally reconstructed geometry and gravitational constitutive physics.

The response determines

$$\boxed{\Theta \implies e \implies g, B = e \wedge e} \quad (277)$$

The infinitesimal closed-response criterion determines the torsion-free sector,

$$\boxed{W_\gamma(P) = 1 + O(\ell^3) \quad \forall P, \gamma \iff T^a = 0} \quad (278)$$

under the assumptions of Sec. 2. Lorentz covariance restricts the constitutive tensor to

$$\boxed{\mathfrak{C} = \alpha I + \beta \star} \quad (279)$$

But the spinless torsion-free bulk theory does not distinguish the identity/Holst direction, and none of the operational structures fixes the magnitude of the Hodge-dual coefficient.

The complete logical statement is therefore

$$\boxed{\begin{array}{l} \text{operational response} \implies e, g, B, \\ \text{closed-response triviality} \implies T = 0, \\ \text{Lorentz covariance} \implies \mathfrak{C} = \alpha I + \beta \star, \\ \text{spinless torsion-free reduction} \implies \text{Einstein classical equivalence class,} \\ \mathfrak{A}_{\text{op}} \not\Rightarrow G, \gamma, \Lambda. \end{array}} \quad (280)$$

The ambiguity in γ is dormant in the present sector because the Holst direction does not alter the torsion-free classical bulk equations. It becomes potentially observable only when the connection is probed by matter carrying intrinsic Lorentz spin. The next section therefore turns to the spinful sector and studies how the constitutive map $\mathfrak{C}$ enters the algebraic spin–torsion response measured by the closed-response phase.

6 Spin as an Operational Probe of Constitutive Gravity

The preceding sections separated two structures that should not be identified. Operational soldering determines the local Lorentzian geometry and the simple area bivector

$$B^{ab} = e^a \wedge e^b, \quad (281)$$

whereas Lorentz covariance restricts, but does not uniquely determine, the local linear constitutive map

$$H_{ab} = \mathfrak{C}(B)_{ab}, \quad \mathfrak{C} = \alpha I + \beta \star. \quad (282)$$

In the spinless torsion-free bulk sector, the relative coefficient α/β is not resolved by the classical field equations. The purpose of the present section is to show that intrinsic spin changes this conclusion in a precise way.

The main point is not that spin sources torsion; that is a standard feature of first-order gravitational theories [?, ?]. Rather, within the present operational framework, spin-generated torsion has an immediate closed-response interpretation: it changes the area-order phase of an infinitesimal response loop. Consequently, the constitutive freedom that is invisible in the spinless torsion-free sector can become visible to operational closed-response observables once a nonzero spin current is present.

6.1 Lorentz spin current and the connection equation

Let the matter fields be collectively denoted by ψ , and suppose that their first-order action depends on the coframe and Lorentz connection,

$$S_m = S_m[e, \omega, \psi]. \quad (283)$$

We define the Lorentz spin current three-form by the connection variation. With the sign convention used throughout this section,

$$\delta_\omega S_m = -\frac{1}{2} \int \delta\omega^{ab} \wedge \tau_{ab}. \quad (284)$$

Equivalently, up to the corresponding variational sign convention,

$$\tau_{ab} = 2 \frac{\delta S_m}{\delta \omega^{ab}}. \quad (285)$$

Consider the first-order gravitational action

$$S_g = \frac{1}{2} \int H_{ab} \wedge R^{ab} - \Lambda \int \text{vol}_e, \quad (286)$$

where

$$H_{ab} = \mathfrak{C}(B)_{ab} = (\alpha I + \beta \star) B_{ab}. \quad (287)$$

The coframe is held fixed in the connection variation, so that

$$\delta_\omega R^{ab} = D\delta\omega^{ab}. \quad (288)$$

Neglecting the corresponding boundary term,

$$\delta_\omega S_g = -\frac{1}{2} \int DH_{ab} \wedge \delta\omega^{ab}. \quad (289)$$

With the convention (284), the connection equation is therefore

$$\boxed{DH_{ab} = \tau_{ab}} \quad (290)$$

Equation (290) is the constitutive connection equation relevant to the present analysis. It should not be interpreted as a new field equation independent of first-order gravity. Its role here is to display explicitly how the operationally reconstructed area variable B , the constitutive map $\mathfrak{C}$, and the matter spin current enter the same local response law.

6.2 The algebraic spin–torsion map

Because the Lorentz connection is metric compatible and preserves the internal orientation,

$$D\star = \star D \quad (291)$$

on internal bivectors. Hence

$$DH = \mathfrak{C}(DB). \quad (292)$$

For

$$B^{ab} = e^a \wedge e^b, \quad (293)$$

one obtains

$$\begin{aligned} DB^{ab} &= D(e^a \wedge e^b) \\ &= De^a \wedge e^b - e^a \wedge De^b \\ &= \boxed{T^a \wedge e^b - e^a \wedge T^b} \end{aligned} \quad (294)$$

It is useful to define the coframe-dependent linear map

$$\mathcal{T}_e : \Omega^2(M, V) \longrightarrow \Omega^3(M, \Lambda^2 V) \quad (295)$$

by

$$\boxed{[\mathcal{T}_e(T)]^{ab} := T^a \wedge e^b - e^a \wedge T^b} \quad (296)$$

For a nondegenerate four-dimensional coframe, $\mathcal{T}_e$ is an algebraic isomorphism between the 24 independent torsion components and the corresponding 24 components of DB^{ab} . An explicit component proof is given in Appendix D.

The connection equation therefore becomes

$$\boxed{(\alpha I + \beta \star) \mathcal{T}_e(T) = \tau} \quad (297)$$

For real coefficients,

$$\alpha^2 + \beta^2 \neq 0 \quad (298)$$

is sufficient for the constitutive map to be invertible, since

$$\boxed{(\alpha I + \beta \star)^{-1} = \frac{\alpha I - \beta \star}{\alpha^2 + \beta^2}} \quad (299)$$

Here we have used

$$\star^2 = -I. \quad (300)$$

Consequently,

$$\boxed{T = \mathcal{T}_e^{-1} \left[\frac{\alpha I - \beta \star}{\alpha^2 + \beta^2} \tau \right]} \quad (301)$$

Theorem 6.1 (Algebraic Spin–Torsion Response). *Let e be a nondegenerate coframe on an oriented four-dimensional Lorentzian manifold, and let*

$$\mathfrak{C} = \alpha I + \beta \star, \quad \alpha^2 + \beta^2 \neq 0, \quad (302)$$

be a real local linear Lorentz-equivariant constitutive map. In the nonpropagating connection sector governed by

$$DH = \tau, \quad H = \mathfrak{C}(B), \quad B = e \wedge e, \quad (303)$$

the matter spin current determines the torsion uniquely and algebraically:

$$\boxed{T = \mathcal{T}_e^{-1} \mathfrak{C}^{-1}(\tau)} \quad (304)$$

In particular,

$$\tau = 0 \iff T = 0 \quad (305)$$

within this nondegenerate algebraic branch.

Proof. Equation (290), together with $D\mathfrak{C} = \mathfrak{C}D$, gives

$$\mathfrak{C}(DB) = \tau. \quad (306)$$

Using

$$DB = \mathcal{T}_e(T), \quad (307)$$

we obtain

$$\mathfrak{C}\mathcal{T}_e(T) = \tau. \quad (308)$$

Both $\mathfrak{C}$ and $\mathcal{T}_e$ are invertible under the stated assumptions, which yields Eq. (304). $\square$

The theorem clarifies the logical status of the torsion-free branch used earlier. Vanishing torsion is not implied by the compact motion–staticity state alone. In the algebraic connection branch it is instead the spinless member

$$\tau = 0 \quad (309)$$

of the more general constitutive response

$$\tau \mapsto T[\tau; \alpha, \beta, e]. \quad (310)$$

6.3 Closed-response phase generated by spin

Section 2 showed that for a sufficiently small contractible loop

$$\gamma = \partial\Sigma \quad (311)$$

and a Lorentz-parallel-transported generator P ,

$$\log W_\gamma(P) = -\frac{i}{\hbar} \int_\Sigma P_a T^a + O(\ell^3). \quad (312)$$

Substitution of Eq. (301) gives

$$\boxed{\log W_\gamma(P) = -\frac{i}{\hbar} \int_\Sigma P_a \left\{ \mathcal{T}_e^{-1} \left[\frac{\alpha I - \beta \star}{\alpha^2 + \beta^2} \tau \right] \right\}^a + O(\ell^3)} \quad (313)$$

This equation provides the direct operational chain

$$\boxed{\tau \longrightarrow T \longrightarrow \arg W_\gamma} \quad (314)$$

Proposition 6.2 (Spin-Generated Response-Loop Phase). *Under the hypotheses of Theorem 6.1, a nonzero spin current produces a unique local torsion field. For every point at which this torsion is nonzero, there exist a sufficiently small oriented two-surface Σ and a generator covector P from any shell-spanning family such that*

$$\log W_{\partial\Sigma}(P) = O(\ell^2) \quad (315)$$

has a nonvanishing area-order coefficient. Thus spin-generated torsion is operationally detectable by the closed-response family.

Proof. If $T \neq 0$ at a point x , then there exist tangent vectors X, Y for which the internal vector

$$T(X, Y) \in V_x \quad (316)$$

is nonzero. Since the chosen massive generator family spans V_x^* , there exists P such that

$$P(T(X, Y)) \neq 0. \quad (317)$$

Choosing a sufficiently small surface whose oriented tangent plane at x is spanned by X, Y , the area-order term in Eq. (312) is nonzero. $\square$

This result should be contrasted with the spinless response-loop criterion. In the spinless branch,

$$\tau = 0 \implies T = 0 \implies W_\gamma(P) = 1 + O(\ell^3). \quad (318)$$

With intrinsic spin,

$$\tau \neq 0 \implies T \neq 0 \quad (319)$$

generically produces an area-order phase. Thus response-loop triviality should not be imposed universally once spinful matter is admitted. It is instead the operational characterization of the torsion-free spinless sector.

6.4 Constitutive magnitude and constitutive orientation

The spin–torsion equation also separates two conceptually different pieces of the constitutive map. Write

$$\mathfrak{C} = \beta(\star + \chi I), \quad \chi := \frac{\alpha}{\beta}, \quad \beta \neq 0. \quad (320)$$

Then

$$\mathfrak{C}^{-1} = \frac{1}{\beta} \frac{\chi I - \star}{1 + \chi^2}. \quad (321)$$

Consequently,

$$T = \frac{1}{\beta} \mathcal{T}_e^{-1} \left[\frac{\chi I - \star}{1 + \chi^2} \tau \right]. \quad (322)$$

Equation (322) makes two independent response parameters visible. The absolute coefficient

$$|\beta|^{-1} \quad (323)$$

sets the overall magnitude of the spin–torsion response, whereas

$$\boxed{\chi = \frac{\alpha}{\beta}} \quad (324)$$

controls the relative action of the identity and Hodge-dual constitutive channels.

In the conventional Einstein–Holst parametrization

$$\mathfrak{C} = \frac{1}{2\kappa} \left(\star + \frac{1}{\gamma} I \right), \quad (325)$$

one has

$$\beta = \frac{1}{2\kappa}, \quad \chi = \frac{1}{\gamma}. \quad (326)$$

Thus the two types of constitutive information may be summarized as

κ^{-1} : overall gravitational response stiffness, γ^{-1} : relative constitutive orientation.

(327)

The second quantity is invisible in the classical spinless torsion-free bulk sector discussed in Section 5, but it enters Eq. (322) whenever the spin current is nonzero. Accordingly, two first-order constitutive theories sharing the same operational soldering data

$$\Theta, \quad e, \quad g, \quad B \quad (328)$$

can nevertheless generate different closed-response phases in the presence of identical spin sources if their constitutive ratios α/β differ.

This yields the following refinement of the constitutive underdetermination result.

Proposition 6.3 (Spin Resolves Constitutive Directions). *Let*

$$\mathfrak{C}_1 = \alpha_1 I + \beta_1 \star, \quad \mathfrak{C}_2 = \alpha_2 I + \beta_2 \star \quad (329)$$

be two nondegenerate real Lorentz-equivariant constitutive maps on the same operationally soldered geometry. If

$$\mathfrak{C}_1^{-1} \neq \mathfrak{C}_2^{-1}, \quad (330)$$

then there exist spin currents τ for which

$$T_1[\tau] \neq T_2[\tau]. \quad (331)$$

For shell-spanning operational generators, the corresponding infinitesimal closed-response families distinguish these torsion responses at area order.

The proposition does not imply that every physical matter sector distinguishes every pair of constitutive maps. A restricted spin current may occupy only part of the algebraic source space. The statement is instead that the constitutive ambiguity that is degenerate in the torsion-free spinless theory is no longer operationally redundant once sufficiently rich spin sources are available.

6.5 Dirac matter as an illustrative spin source

The simplest standard example is a Dirac field. Define the axial current

$$J_5^a := \bar{\psi} \gamma^a \gamma_5 \psi. \quad (332)$$

For minimally coupled Dirac matter, the Lorentz spin current occupies the axial spin sector. Introducing the coframe three-form

$$\eta_a := \frac{1}{3!} \epsilon_{abcd} e^b \wedge e^c \wedge e^d, \quad (333)$$

its algebraic structure may be written schematically as

$$\tau_{ab}^{(D)} \propto \epsilon_{ab}{}^{cd} J_{5c} \eta_d \quad (334)$$

where the numerical coefficient depends on the normalization conventions for the Dirac and gravitational actions.

For the pure Einstein–Cartan direction,

$$\alpha = 0, \quad (335)$$

Eq. (301) gives the familiar purely axial, totally antisymmetric torsion structure,

$$T_{\text{EC}}^a = C_D \kappa \epsilon^a{}_{bcd} J_5^d e^b \wedge e^c \quad (336)$$

where C_D is fixed once the action and ϵ -tensor conventions are chosen. The explicit coefficient used in this paper is derived in Appendix D.

For the general Einstein–Holst constitutive map, the inverse constitutive operator mixes the two Lorentz-equivariant bivector channels. The resulting torsion therefore has the general form

$$T_D^a = A(\kappa, \gamma) \epsilon^a{}_{bcd} J_5^d e^b \wedge e^c + B(\kappa, \gamma) J_{5b} e^a \wedge e^b \quad (337)$$

with coefficients A and B determined algebraically by Eq. (322). Their exact expressions depend on the adopted normalization and on whether nonminimal fermionic couplings are included; these convention issues are collected in Appendix D. In the Einstein–Cartan limit the second structure disappears and Eq. (336) is recovered.

The corresponding response-loop phase is

$$\begin{aligned} \log W_\gamma(P) &= -\frac{i}{\hbar} \int_\Sigma P_a T_D^a + O(\ell^3) \\ &= -\frac{i}{\hbar} \int_\Sigma \left[A(\kappa, \gamma) P_a \epsilon^a{}_{bcd} J_5^d e^b \wedge e^c \right. \\ &\quad \left. + B(\kappa, \gamma) P_a J_{5b} e^a \wedge e^b \right] + O(\ell^3). \end{aligned} \quad (338)$$

For an infinitesimal surface with oriented area bivector Σ^{bc} , this may be written locally as

$$\arg W_\gamma(P) = -\frac{1}{\hbar} \left[A(\kappa, \gamma) P_a \epsilon^a{}_{bcd} J_5^d + B(\kappa, \gamma) P^{[b} J_5^{c]} \right] \Sigma_{bc} + O(\ell^3) \quad (339)$$

Equation (339) is the operational content of the example. The Dirac field first produces a Lorentz spin current, the constitutive map converts that current into torsion, and the torsion is then measured by the translational closed-response phase:

$$J_5 \longrightarrow \tau \longrightarrow T \longrightarrow \arg W_\gamma \quad (340)$$

Integrating out the algebraic torsion produces the familiar local four-fermion interactions of Einstein–Cartan/Holst gravity. The dependence of those interactions on the Holst/Immirzi sector, and its sensitivity to the choice of minimal or nonminimal fermion coupling, is well known [?, ?, ?]. The present construction does not claim a new fermionic phenomenology. Its role is to identify the same constitutive dependence directly at the level of an operational closed-response observable.

6.6 What spin does and does not determine

The spinful extension changes the status of the relative constitutive coefficient but does not remove the constitutive underdetermination established in the preceding section.

First, intrinsic spin can make the ratio

$$\frac{\alpha}{\beta} \quad (341)$$

operationally relevant because different values generally produce different torsion responses to the same spin current.

Second, this does not imply that the operational MS structure fixes that ratio. The matter content itself must supply a spin source, and the constitutive map must still be specified before Eq. (301) determines the torsion.

Third, neither the existence of spin nor the response-loop measurement fixes the absolute constitutive scale. A common rescaling

$$\mathfrak{C} \mapsto \lambda \mathfrak{C} \quad (342)$$

changes the torsion response to a fixed spin current while leaving the underlying kinematical soldering data unchanged. Thus the numerical gravitational stiffness remains independent input.

The logical structure of the spinful sector is therefore

$$\begin{aligned} \Theta &\implies e, g, B, \\ \mathfrak{C} &= \alpha I + \beta \star, \\ \tau &\implies T = \mathcal{T}_e^{-1} \mathfrak{C}^{-1}(\tau), \\ T &\implies \arg W_\gamma \quad \text{at area order,} \end{aligned} \quad (343)$$

spinful closed response $\implies$ possible sensitivity to α/β ,

$\Theta, W, \tau \not\Rightarrow \alpha, \beta$ without an additional constitutive law.

This distinction is central to the interpretation adopted here. Spin does not derive the gravitational constitutive law; it probes it. The spinless theory is insensitive to part of the constitutive freedom because torsion vanishes, whereas spinful matter provides a source with which that freedom can be operationally interrogated.

The next section asks a different question. Rather than using matter spin to probe an already chosen constitutive map, we investigate whether an additional symmetry of the area–excitation relation itself can restrict the two-parameter Lorentz-equivariant family

$$\mathfrak{C} = \alpha I + \beta \star. \quad (344)$$

We shall show that invariance under a specific pure constitutive reciprocity implies a closure relation

$$\mathfrak{C}^2 \propto -I \quad (345)$$

and conditionally selects the Hodge-dual Einstein–Cartan direction. That reciprocity principle, however, will remain an additional constitutive axiom rather than a consequence of the operational soldering or of the spin response derived above.

7 Constitutive Reciprocity and Conditional Selection of the Einstein Direction

The preceding sections separated two logically distinct structures. The operational response determines the solder form and hence the simple area bivector

$$B^{ab} = e^a \wedge e^b, \quad (346)$$

whereas local linear Lorentz covariance restricts the constitutive relation between geometric area and gravitational excitation to

$$\boxed{H = \mathfrak{C}B, \quad \mathfrak{C} = \alpha I + \beta \star} \quad (347)$$

Here

$$\star : \Lambda^2 V \longrightarrow \Lambda^2 V \quad (348)$$

is the internal Hodge operator associated with the oriented Lorentzian internal metric, and

$$\boxed{\star^2 = -I} \quad (349)$$

on real internal bivectors.

Nothing established so far fixes the ratio

$$\frac{\alpha}{\beta} \quad (350)$$

or the overall scale of $\mathfrak{C}$. In particular, operational soldering, the response-loop torsion criterion, and Lorentz covariance alone do not select the Hodge-dual direction from the two-dimensional constitutive algebra $\text{span}_{\mathbb{R}}\{I, \star\}$.

In this section we ask a conditional question. Suppose that, in addition to the operational principles already introduced, the area–excitation pair

$$(B, H) \quad (351)$$

is required to possess a reciprocity symmetry exchanging geometric area and gravitational excitation. We show that a particularly simple reciprocity postulate imposes a closure relation on the constitutive map and, in Lorentzian signature, selects the Hodge-dual Einstein–Cartan direction up to an overall scale and sign.

The logical status of this assumption is essential: the reciprocity principle introduced below is *not* derived from the compact motion–staticity state, from the soldering response Θ , or from the response-loop construction. It is an additional constitutive axiom.

7.1 Area–excitation reciprocity

The bivector B and the excitation H have different physical dimensions in general. We therefore introduce a nonzero real conversion parameter

$$\zeta \neq 0 \quad (352)$$

and define the linear transformation

$$\boxed{\mathcal{R}_{\zeta} : (B, H) \longmapsto (B', H') = (\zeta H, -\zeta^{-1} B)} \quad (353)$$

Applying the transformation twice gives

$$\mathcal{R}_\zeta^2(B, H) = (-B, -H), \quad (354)$$

and hence

$$\boxed{\mathcal{R}_\zeta^2 = -I, \quad \mathcal{R}_\zeta^4 = I} \quad (355)$$

Thus the reciprocity operation is a quarter-turn on the doubled constitutive space

$$\Lambda^2 V \oplus \Lambda^2 V, \quad (356)$$

rather than an involutive exchange.

We now require that the constitutive graph

$$\Gamma_\mathfrak{C} := \{(B, H) : H = \mathfrak{C}B\} \quad (357)$$

be invariant under $\mathcal{R}_\zeta$. Explicitly,

$$(B, H) \in \Gamma_\mathfrak{C} \implies \mathcal{R}_\zeta(B, H) \in \Gamma_\mathfrak{C}. \quad (358)$$

This is the constitutive reciprocity assumption used below.

7.2 Constitutive closure

Graph invariance immediately imposes a quadratic condition on $\mathfrak{C}$.

Proposition 7.1 (Reciprocity implies constitutive closure). *Let*

$$H = \mathfrak{C}B$$

be a real linear constitutive law on $\Lambda^2 V$, and let $\mathcal{R}_\zeta$ be defined by Eq. (353). If the graph $\Gamma_\mathfrak{C}$ is invariant under $\mathcal{R}_\zeta$, then

$$\boxed{\mathfrak{C}^2 = -\zeta^{-2}I} \quad (359)$$

Proof. Take an arbitrary

$$B \in \Lambda^2 V$$

and set

$$H = \mathfrak{C}B.$$

Under reciprocity,

$$B' = \zeta H = \zeta \mathfrak{C}B, \quad (360)$$

while

$$H' = -\zeta^{-1}B. \quad (361)$$

Graph invariance requires

$$H' = \mathfrak{C}B'. \quad (362)$$

Substituting Eqs. (360) and (361),

$$-\zeta^{-1}B = \mathfrak{C}(\zeta \mathfrak{C}B) = \zeta \mathfrak{C}^2 B. \quad (363)$$

Since B is arbitrary and $\zeta \neq 0$,

$$\mathfrak{C}^2 = -\zeta^{-2}I, \quad (364)$$

as claimed. $\square$

It is useful to introduce the dimensionless normalized constitutive operator

$$\boxed{J := \zeta \mathfrak{C}} \quad (365)$$

Then Eq. (359) becomes

$$\boxed{J^2 = -I} \quad (366)$$

The reciprocity-invariant constitutive law therefore defines a complex structure on the real six-dimensional bivector space.

7.3 Reciprocity-closure rigidity

We now combine the closure relation with the Lorentz-equivariant classification derived in Sec. 4. Recall that

$$\mathfrak{C} = \alpha I + \beta \star, \quad (367)$$

with real α, β . Using $\star^2 = -I$, one obtains

$$\begin{aligned} \mathfrak{C}^2 &= (\alpha I + \beta \star)^2 \\ &= \alpha^2 I + 2\alpha\beta \star + \beta^2 \star^2 \\ &= \boxed{(\alpha^2 - \beta^2)I + 2\alpha\beta \star} \end{aligned} \quad (368)$$

Comparison with

$$\mathfrak{C}^2 = -\zeta^{-2} I \quad (369)$$

gives the two real algebraic conditions

$$2\alpha\beta = 0, \quad (370)$$

$$\alpha^2 - \beta^2 = -\zeta^{-2}. \quad (371)$$

The resulting rigidity statement is the main theorem of this section.

Theorem 7.2 (Constitutive Reciprocity–Closure Theorem). *Let V be an oriented four-dimensional real Lorentz vector space, and let*

$$\mathfrak{C} : \Lambda^2 V \longrightarrow \Lambda^2 V$$

be a real linear $SO^+(1, 3)$ -equivariant constitutive map. Assume:

1. *$\mathfrak{C}$ relates area and excitation by*

$$H = \mathfrak{C}B;$$

2. *the constitutive graph is invariant under the pure reciprocity transformation*

$$(B, H) \longmapsto (\zeta H, -\zeta^{-1} B), \quad \zeta \in \mathbb{R} \setminus \{0\}.$$

Then

$$\boxed{\mathfrak{C} = \pm \zeta^{-1} \star} \quad (372)$$

Thus pure constitutive reciprocity uniquely selects the Hodge-dual direction up to an overall real scale and sign.

Proof. Lorentz equivariance implies

$$\mathfrak{G} = \alpha I + \beta \star . \quad (373)$$

By Proposition 7.1, graph invariance gives

$$\mathfrak{G}^2 = -\zeta^{-2} I. \quad (374)$$

Equations (370) and (371) therefore hold.

From

$$2\alpha\beta = 0,$$

either $\alpha = 0$ or $\beta = 0$.

If $\beta = 0$, then Eq. (371) would require

$$\alpha^2 = -\zeta^{-2}, \quad (375)$$

which has no solution for real α and real nonzero ζ .

Hence

$$\alpha = 0. \quad (376)$$

Eq. (371) then gives

$$-\beta^2 = -\zeta^{-2}, \quad (377)$$

so that

$$\beta = \pm\zeta^{-1}. \quad (378)$$

Therefore

$$\mathfrak{G} = \pm\zeta^{-1} \star . \quad (379)$$

□

The theorem may equivalently be stated as follows: among real Lorentz-equivariant endomorphisms of the bivector space, the only ones whose normalized form defines the complex structure required by pure area–excitation reciprocity are

$$\boxed{J = \pm \star} \quad (380)$$

7.4 Relation to the Einstein–Holst parameterization

To compare the result with the conventional first-order parameterization, write

$$\boxed{\mathfrak{G} = \frac{1}{2\kappa} \left(\star + \frac{1}{\gamma} I \right)} \quad (381)$$

where the precise normalization of κ and sign of γ depend on the action conventions adopted in Appendix C.

Comparison with Theorem 7.2 immediately gives:

Corollary 7.3 (Removal of the Holst direction under pure reciprocity). *Within the convention (381), pure constitutive reciprocity requires*

$$\boxed{\frac{1}{\gamma} = 0} \quad (382)$$

or equivalently

$$\boxed{|\gamma| = \infty} \quad (383)$$

The constitutive map therefore reduces to the Einstein–Cartan/Hodge-dual direction.

This conclusion concerns the off-shell constitutive law. As shown in Sec. 5, the identity/Holst contribution is already invisible in the torsion-free spinless classical bulk equations. The reciprocity condition is therefore stronger than what is needed to recover the classical Einstein equations in that sector. Its effect is to select a particular representative from the wider Einstein-equivalent first-order constitutive family.

When spinful matter is present, however, the distinction between $\alpha = 0$ and $\alpha \neq 0$ can enter the algebraic torsion response. The reciprocity selection therefore has genuine constitutive content beyond the purely torsion-free classical bulk theory.

7.5 Why the result is conditional

It is important not to reverse the logical direction of Theorem 7.2. The theorem establishes

$$\boxed{\text{Lorentz-equivariant constitutive law} + \text{pure reciprocity} \implies \mathfrak{C} \propto \star} \quad (384)$$

It does *not* establish

$$\boxed{\text{MS state} + \Theta + W \implies \text{pure reciprocity}} \quad (385)$$

Indeed, the operational structures developed earlier determine

$$e, \quad g, \quad B = e \wedge e, \quad (386)$$

and the response-loop construction supplies an operational diagnostic of torsion. None of these structures introduces an independent excitation H or a symmetry exchanging B with H . Such an exchange relates a kinematically reconstructed geometric quantity to a dynamical constitutive response and therefore contains new physical information.

In particular, ordinary reversal of the local response phase,

$$W^{-1} = W^\dagger, \quad (387)$$

does not provide the reciprocity (353). The former is an involutive reversal operation, whereas the latter satisfies

$$\mathcal{R}_\zeta^2 = -I. \quad (388)$$

Thus the complex-structure closure required here does not follow merely from unitarity or reversibility of the local response.

Accordingly, reciprocity should be regarded as a candidate constitutive selection principle rather than as an already derived MS axiom.

7.6 Generalized closure does not remove the identity component

The strength of the preceding theorem depends on the *pure* closure relation

$$\mathfrak{C}^2 = -\zeta^{-2}I. \quad (389)$$

A shifted or generalized closure condition need not select the same constitutive direction.

For example, consider

$$(\mathfrak{C} - \alpha_0 I)^2 = -\lambda^2 I \quad (390)$$

with real α_0 and $\lambda \neq 0$. Then the constitutive map

$$\mathfrak{C} = \alpha_0 I \pm \lambda \star \quad (391)$$

satisfies Eq. (390). The identity component therefore survives.

Hence

$$\boxed{\text{generalized closure} \not\Rightarrow \alpha = 0} \quad (392)$$

The elimination of the Holst direction is a consequence of the specific pure reciprocity transformation (353), not of an arbitrary quadratic constitutive relation.

This distinction is important when comparing different closure conditions in the literature.

7.7 Lorentzian Hodge duality as the constitutive complex structure

The normalized constitutive operator

$$J = \zeta \mathfrak{C} \quad (393)$$

satisfies

$$J^2 = -I \quad (394)$$

under reciprocity closure. By Theorem 7.2, Lorentz covariance then gives

$$\boxed{J = \pm \star} \quad (395)$$

Thus the complex structure selected by the constitutive reciprocity is the same real complex structure already provided on bivectors by the oriented Lorentzian metric.

This observation provides a useful geometric interpretation of the selection theorem:

$$\boxed{\text{constitutive quarter-turn} = \text{Lorentzian Hodge quarter-turn}} \quad (396)$$

up to orientation sign and constitutive scale.

The result also depends on Lorentzian signature. In four-dimensional Euclidean signature,

$$\star^2 = +I \quad (397)$$

on real two-forms. A real map

$$\mathfrak{C} = \alpha I + \beta \star \quad (398)$$

then satisfies

$$\mathfrak{C}^2 = (\alpha^2 + \beta^2)I + 2\alpha\beta\star, \quad (399)$$

which cannot equal $-\zeta^{-2}I$ for real α, β, ζ . Pure real reciprocity closure therefore fits naturally with the Lorentzian bivector complex structure.

In the present framework this should not be interpreted as an independent derivation of Lorentzian spacetime signature. The Lorentzian internal metric has already been reconstructed before the constitutive sector is introduced. The observation is instead a consistency relation between the previously reconstructed signature and the possible real reciprocity-closed constitutive law.

7.8 Reciprocity does not determine the gravitational scale

Although reciprocity removes the relative identity/Holst direction, it does not fix the magnitude of the remaining constitutive map. The theorem gives

$$\mathfrak{C} = \pm \zeta^{-1} \star, \quad (400)$$

but leaves

$$|\zeta| \quad (401)$$

arbitrary.

For the Einstein-normalized constitutive law,

$$H_{ab} = -\frac{1}{2\kappa}(\star B)_{ab}, \quad (402)$$

one therefore identifies, up to the sign and normalization conventions,

$$|\zeta|^{-1} \sim \frac{1}{\kappa} = \frac{c^4}{8\pi G} \quad (403)$$

Consequently,

$$\boxed{\text{reciprocity closure fixes the constitutive direction, not the constitutive stiffness}} \quad (404)$$

In particular,

$$\boxed{\text{pure reciprocity} \not\Rightarrow G} \quad (405)$$

This agrees with the constitutive-underdetermination result of Sec. 5. The compact state geometry, operational soldering, and closed response determine the kinematical geometry and constrain the tensor form of the gravitational response, but the overall conversion between curvature excitation and operational area remains an independent physical scale.

7.9 Relation to reciprocity and closure in premetric field theory

The algebraic structure above has a close analogue in premetric electrodynamics. There one distinguishes the field strength from its excitation and asks whether an electric–magnetic reciprocity transformation is compatible with a local linear constitutive law. Reciprocity and closure conditions lead, under suitable assumptions, to a constitutive complex structure and to the emergence of a Hodge-type operator [?, ?].

The present construction should be compared with that literature at the level of constitutive logic rather than identified with it. Here the pair being exchanged is

$$(B, H), \quad (406)$$

where $B = e \wedge e$ has already been reconstructed from the operational soldering response, while H is the curvature excitation of the first-order gravitational sector. The operational origin of B is therefore specific to the present reconstruction.

No claim is made that the closure algebra

$$\mathfrak{C}^2 \propto -I \quad (407)$$

is new in itself. The result established here is the conditional placement of that algebra inside the operational reconstruction:

$$\boxed{\Theta \implies B = e \wedge e, \quad \text{Lorentz covariance} \implies \mathfrak{C} = \alpha I + \beta \star, \quad \text{reciprocity} \implies \mathfrak{C} \propto \star} \quad (408)$$

7.10 Logical endpoint

Combining the results of the present and preceding sections gives the following hierarchy:

$$\begin{aligned}
\Theta &\implies e, g, \\
\Theta &\implies B = e \wedge e, \\
\text{response loops} &\implies \text{operational torsion diagnostic}, \\
\text{Lorentz covariance} &\implies \mathfrak{C} = \alpha I + \beta \star, \\
\text{spinless torsion-free bulk} &\implies \text{Einstein equivalence class}, \\
\text{pure constitutive reciprocity} &\implies \mathfrak{C} = \pm \zeta^{-1} \star, \\
\text{operational principles} &\not\Rightarrow |\zeta|^{-1} \sim G^{-1}.
\end{aligned} \tag{409}$$

The reciprocity principle therefore supplies a possible *conditional* selection of the Einstein–Cartan constitutive direction, but it does not close the entire gravitational theory. The remaining scale is precisely the constitutive normalization already identified with gravitational response stiffness.

The distinction between

$$\boxed{\text{kinematical reconstruction}} \quad \text{and} \quad \boxed{\text{constitutive calibration}} \tag{410}$$

is therefore preserved even after reciprocity is imposed.

This is the appropriate endpoint of the constitutive analysis: the operational response determines the geometric variables on which gravitational dynamics acts, Lorentz symmetry sharply restricts the admissible lowest-order linear response law, and an additional reciprocity principle can select its Hodge-dual orientation. Neither the reciprocity condition nor the preceding operational construction determines the absolute strength of gravity.

A Covariant Small-Loop Expansion of the Response Phase

This appendix gives a more detailed derivation of the infinitesimal closed-response expansion used in Sec. 2. Particular attention is paid to two points that are suppressed in the main text: the extension of the base-point generator from the boundary of the loop to the spanning surface, and the order at which Lorentz curvature enters the scalar translational response.

Throughout this appendix, $x \in M$ is fixed,

$$\gamma_\ell = \partial \Sigma_\ell$$

is a smooth family of contractible loops shrinking to x , and

$$\text{length}(\gamma_\ell) = O(\ell), \quad \text{area}(\Sigma_\ell) = O(\ell^2).$$

The solder form is assumed to be C^2 , while the Lorentz connection is C^1 .

A.1 A local Lorentz gauge adapted to the base point

Let

$$\omega^a{}_b = \omega^a{}_{b\mu} dx^\mu$$

be a smooth Lorentz connection in a neighborhood of x .

Lemma A.1 (Base-point Lorentz gauge). *There exists a smooth local Lorentz gauge in which*

$$\boxed{\omega^a_{b\mu}(x) = 0} \quad (411)$$

at the chosen base point. In this gauge,

$$\omega^a_{b\mu}(y) = O(|y - x|) \quad (412)$$

for y in a sufficiently small neighborhood of x .

Proof. The value of a connection one-form at one point is gauge dependent. A local Lorentz transformation may therefore be chosen so that the transformed connection vanishes at the specified point. Smoothness then implies the first-order estimate

$$\omega^a_{b\mu}(y) = \partial_\nu \omega^a_{b\mu}(x)(y^\nu - x^\nu) + O(|y - x|^2),$$

and hence

$$\omega(y) = O(|y - x|).$$

□

For a shrinking loop contained in a coordinate ball of radius $O(\ell)$, this gives

$$\boxed{\omega|_{\gamma_\ell} = O(\ell), \quad \omega|_{\Sigma_\ell} = O(\ell)} \quad (413)$$

A.2 Parallel transport of the generator

Let $P_a(s)$ satisfy the dual parallel-transport equation

$$\frac{dP_a}{ds} - \dot{\gamma}^\mu \omega^b_{a\mu} P_b = 0, \quad P_a(0) = P_a. \quad (414)$$

Choose the parametrization of the shrinking loop such that

$$\dot{\gamma}_\ell^\mu(s) = O(\ell)$$

uniformly for $s \in [0, 1]$.

Lemma A.2 (Uniform parallel-generator estimate). *In the gauge (411),*

$$\boxed{P_a(s) = P_a + O(\ell^2)} \quad (415)$$

uniformly along the shrinking loop.

Proof. Using

$$\dot{\gamma} = O(\ell), \quad \omega(\gamma_\ell) = O(\ell),$$

Eq. (414) gives

$$\frac{dP_a}{ds} = O(\ell^2).$$

Integration over a parameter interval of $O(1)$ yields

$$P_a(s) - P_a(0) = O(\ell^2).$$

□

In particular, the total Lorentz holonomy satisfies

$$P_a(1) - P_a(0) = O(\ell^2). \quad (416)$$

More precisely, its leading coefficient is controlled by the Lorentz curvature,

$$\Delta_\gamma P = O(R \ell^2). \quad (417)$$

A.3 Surface extension of the base-point generator

The expression

$$\int_{\Sigma_\ell} P_a T^a$$

used in a small-loop formula requires a convention because the parallel-transported generator has been defined canonically only along the boundary path.

Let

$$\widetilde{P}_a(y)$$

be any smooth extension to a neighborhood of x satisfying

$$\widetilde{P}_a(x) = P_a. \quad (418)$$

Lemma A.3 (Extension independence to area order). *Let $\widetilde{P}$ and $\widetilde{P}'$ be two smooth extensions satisfying*

$$\widetilde{P}(x) = \widetilde{P}'(x) = P.$$

Then

$$\boxed{\int_{\Sigma_\ell} (\widetilde{P}_a - \widetilde{P}'_a) T^a = O(\ell^3)} \quad (419)$$

Proof. Since the two extensions agree at x ,

$$\widetilde{P}_a(y) - \widetilde{P}'_a(y) = O(|y - x|) = O(\ell)$$

on the shrinking surface. Since $T^a = O(1)$ and $\text{area}(\Sigma_\ell) = O(\ell^2)$, the integral is $O(\ell^3)$. $\square$

Thus the notation

$$\int_{\Sigma_\ell} P_a T^a \quad (420)$$

is unambiguous through $O(\ell^2)$, provided it is understood as

$$\int_{\Sigma_\ell} \widetilde{P}_a T^a \quad (421)$$

for any smooth extension satisfying $\widetilde{P}(x) = P$.

A.4 Detailed small-loop expansion

We now give the precise result.

Theorem A.4 (Covariant response-loop expansion). *Under the assumptions stated above,*

$$\boxed{I_{\gamma_\ell}(P) = \oint_{\gamma_\ell} P_a(s) e^a = \int_{\Sigma_\ell} \widetilde{P}_a T^a + O(\ell^3)} \quad (422)$$

for any smooth extension $\widetilde{P}$ satisfying $\widetilde{P}(x) = P$.

Equivalently,

$$\boxed{I_{\gamma_\ell}(P) = \frac{1}{2} P_a T^a{}_{\mu\nu}(x) \Sigma_\ell^{\mu\nu} + O(\ell^3)} \quad (423)$$

Proof. Work first in the base-point gauge

$$\omega(x) = 0.$$

By Lemma A.2,

$$\begin{aligned} I_{\gamma_\ell}(P) &= \oint_{\gamma_\ell} [P_a + O(\ell^2)] e^a \\ &= P_a \oint_{\gamma_\ell} e^a + O(\ell^3), \end{aligned} \quad (424)$$

because the line element has total order $O(\ell)$.

Ordinary Stokes' theorem in this fixed local frame gives

$$P_a \oint_{\gamma_\ell} e^a = P_a \int_{\Sigma_\ell} de^a. \quad (425)$$

Using

$$de^a = T^a - \omega^a_b \wedge e^b,$$

we obtain

$$P_a \oint_{\gamma_\ell} e^a = P_a \int_{\Sigma_\ell} T^a - P_a \int_{\Sigma_\ell} \omega^a_b \wedge e^b. \quad (426)$$

On Σ_ℓ ,

$$\omega = O(\ell), \quad e = O(1),$$

and the surface area is $O(\ell^2)$. Hence

$$\int_{\Sigma_\ell} P_a \omega^a_b \wedge e^b = O(\ell^3). \quad (427)$$

Therefore

$$I_{\gamma_\ell}(P) = P_a \int_{\Sigma_\ell} T^a + O(\ell^3). \quad (428)$$

Replacing the constant base-point components P_a by any smooth extension $\tilde{P}_a$ changes the result only by $O(\ell^3)$, by Lemma A.3. This proves Eq. (422).

Finally,

$$T^a = \frac{1}{2} T^a_{\mu\nu}(x) dy^\mu \wedge dy^\nu + O(\ell),$$

so

$$\int_{\Sigma_\ell} \tilde{P}_a T^a = \frac{1}{2} P_a T^a_{\mu\nu}(x) \Sigma_\ell^{\mu\nu} + O(\ell^3),$$

which proves Eq. (423). □

A.5 Gauge independence

Although the proof used a special local Lorentz gauge, the result itself is gauge independent. Under

$$e^a \mapsto \Lambda^a_b e^b, \quad P_a \mapsto P_b (\Lambda^{-1})^b_a,$$

the scalar one-form

$$P_a e^a$$

is invariant. Likewise,

$$T^a \mapsto \Lambda^a_b T^b$$

and hence

$$P_a T^a$$

is invariant.

Therefore both sides of Eq. (422) are Lorentz-gauge scalars.

A.6 Why Lorentz curvature enters one order later

The Lorentz holonomy of the transported generator is itself an area-order effect:

$$\Delta_\gamma P = O(\ell^2).$$

However, its contribution to the scalar response integral is

$$\oint_{\gamma_\ell} (\Delta P)_a e^a = O(\ell^2) O(\ell) = O(\ell^3).$$

Thus

torsion contribution to $I_\gamma = O(\ell^2)$, curvature-induced generator correction = $O(\ell^3)$.
--

(429)

This statement concerns the scalar translational response $I_\gamma(P)$. It does not state that Lorentz curvature itself is a third-order effect: the ordinary Lorentz holonomy remains $O(\ell^2)$.

A.7 Local branch of the phase logarithm

Since

$$I_{\gamma_\ell}(P) = O(\ell^2),$$

there exists, for sufficiently small ℓ , a neighborhood of the identity in which

$$W_{\gamma_\ell}(P) = \exp \left[-\frac{i}{\hbar} I_{\gamma_\ell}(P) \right]$$

has a unique continuous logarithm satisfying

$$\log W_{\gamma_\ell}(P) \longrightarrow 0 \quad (\ell \rightarrow 0).$$

Hence

$\log W_{\gamma_\ell}(P) = -\frac{i}{2\hbar} P_a T^a_{\mu\nu}(x) \Sigma_\ell^{\mu\nu} + O(\ell^3)$
--

(430)

The local logarithm removes any possible ambiguity by integer multiples of $2\pi i$ from the infinitesimal-loop argument.

B The Lorentz Commutant on Real Bivectors

This appendix gives the representation-theoretic details underlying Lemma 4.1.

Let V be an oriented real four-dimensional Lorentz vector space with

$$\eta = \text{diag}(-1, +1, +1, +1).$$

The connected Lorentz group

$$G = SO^+(1, 3)$$

acts naturally on

$$\Lambda^2 V.$$

The internal Hodge operator is

$$(\star X)^{ab} = \frac{1}{2} \epsilon^{ab}_{cd} X^{cd}, \quad X \in \Lambda^2 V. \quad (431)$$

For the Lorentzian signature used here,

$$\boxed{\star^2 = -I} \quad (432)$$

on real bivectors.

B.1 Complex self-dual decomposition

Complexify:

$$\Lambda^2 V_{\mathbb{C}} = \Lambda^2 V \otimes_{\mathbb{R}} \mathbb{C}. \quad (433)$$

Since

$$\star^2 = -I,$$

the complexified Hodge operator has eigenvalues

$$+i, \quad -i.$$

Define

$$P_{\pm} = \frac{1}{2} (I \mp i\star). \quad (434)$$

Then

$$P_{\pm}^2 = P_{\pm}, \quad P_+ P_- = 0, \quad P_+ + P_- = I,$$

and

$$\Lambda^2 V_{\mathbb{C}} = \Lambda_+^2 \oplus \Lambda_-^2, \quad (435)$$

where

$$\Lambda_{\pm}^2 = P_{\pm} \Lambda^2 V_{\mathbb{C}}. \quad (436)$$

Each summand has complex dimension three.

Under the connected Lorentz group, these two spaces are irreducible and inequivalent complex representations. In the conventional $SL(2, \mathbb{C})$ notation they correspond to

$$(1, 0), \quad (0, 1).$$

B.2 Complex commutant

Let

$$A : \Lambda^2 V \longrightarrow \Lambda^2 V$$

be a real linear Lorentz-equivariant map, and let

$$A_{\mathbb{C}}$$

be its complexification.

Because

$$A_{\mathbb{C}}g = gA_{\mathbb{C}} \quad \forall g \in G,$$

Schur's lemma gives

$$A_{\mathbb{C}}|_{\Lambda_+^2} = c_+ I, \quad A_{\mathbb{C}}|_{\Lambda_-^2} = c_- I \quad (437)$$

for complex numbers $c_{\pm}$.

Complex conjugation interchanges Λ_+^2 and Λ_-^2 :

$$\overline{\Lambda_+^2} = \Lambda_-^2. \quad (438)$$

Since A is real,

$$A_{\mathbb{C}}\overline{X} = \overline{A_{\mathbb{C}}X}.$$

Therefore

$$\boxed{c_- = \overline{c_+}} \quad (439)$$

Write

$$c_+ = \alpha + i\beta, \quad \alpha, \beta \in \mathbb{R}.$$

Then

$$c_- = \alpha - i\beta.$$

Now

$$\alpha I + \beta \star$$

acts on Λ_+^2 with eigenvalue

$$\alpha + i\beta$$

and on Λ_-^2 with eigenvalue

$$\alpha - i\beta.$$

Hence

$$A_{\mathbb{C}} = \alpha I + \beta \star. \quad (440)$$

Restriction to the real subspace gives

Theorem B.1 (Real Lorentz bivector commutant).

$$\boxed{\text{End}_{SO^+(1,3)}(\Lambda^2 V) = \mathbb{R}I \oplus \mathbb{R}\star} \quad (441)$$

Proof. The preceding argument proves existence. Uniqueness follows because I and $\star$ are linearly independent over $\mathbb{R}$: if

$$\alpha I + \beta \star = 0,$$

then multiplication by

$$\alpha I - \beta \star$$

gives

$$(\alpha^2 + \beta^2)I = 0,$$

and hence

$$\alpha = \beta = 0.$$

□

B.3 Algebra structure

The commutant is not merely a two-dimensional vector space. Because

$$\star^2 = -I,$$

its multiplication law is

$$(\alpha I + \beta \star)(\alpha' I + \beta' \star) = (\alpha\alpha' - \beta\beta')I + (\alpha\beta' + \beta\alpha') \star. \quad (442)$$

Thus, as a real associative algebra,

$$\boxed{\text{End}_{SO^+(1,3)}(\Lambda^2 V) \simeq \mathbb{C}} \quad (443)$$

with the correspondence

$$I \longleftrightarrow 1, \quad \star \longleftrightarrow i.$$

This observation makes the constitutive parameterization

$$\mathfrak{C} = \alpha I + \beta \star$$

particularly transparent: the identity and Hodge directions are the real and imaginary directions of the Lorentz-equivariant commutant.

B.4 Inverse

For

$$(\alpha, \beta) \neq (0, 0),$$

$$(\alpha I + \beta \star)(\alpha I - \beta \star) = (\alpha^2 + \beta^2)I, \quad (444)$$

and therefore

$$\boxed{(\alpha I + \beta \star)^{-1} = \frac{\alpha I - \beta \star}{\alpha^2 + \beta^2}} \quad (445)$$

In particular, every nonzero real element of the Lorentz bivector commutant is invertible.

C Conventions for Hodge Duality and First-Order Gravity

This appendix fixes the sign and normalization conventions used in the first-order constitutive sector. It is included because factors of two and signs in the Holst and Einstein–Cartan terms vary substantially across the literature.

C.1 Metric and orientation

The internal Lorentz metric is

$$\eta_{ab} = \text{diag}(-1, +1, +1, +1). \quad (446)$$

The orientation is fixed by

$$\epsilon_{0123} = +1. \quad (447)$$

Consequently,

$$\epsilon^{0123} = -1. \quad (448)$$

For an internal bivector $X^{ab} = -X^{ba}$,

$$(\star X)^{ab} := \frac{1}{2} \epsilon^{ab}_{cd} X^{cd}. \quad (449)$$

With the above signature,

$$\boxed{\star^2 = -I} \quad (450)$$

on $\Lambda^2 V$.

C.2 Coframe forms

Let

$$e^a$$

be an oriented orthonormal coframe. We define

$$\eta := \text{vol}_e = \frac{1}{4!} \epsilon_{abcd} e^a \wedge e^b \wedge e^c \wedge e^d, \quad (451)$$

$$\eta_a := \iota_a \eta = \frac{1}{3!} \epsilon_{abcd} e^b \wedge e^c \wedge e^d, \quad (452)$$

$$\eta_{ab} := \iota_b \iota_a \eta = \frac{1}{2} \epsilon_{abcd} e^c \wedge e^d. \quad (453)$$

The operational bivector-valued two-form is

$$\boxed{B^{ab} = e^a \wedge e^b} \quad (454)$$

No factor $1/2$ is included in this definition.

C.3 Connection, curvature, and torsion

The Lorentz connection satisfies

$$\omega_{ab} = -\omega_{ba}.$$

Its curvature and torsion are

$$R^{ab} = d\omega^{ab} + \omega^a_c \wedge \omega^{cb}, \quad (455)$$

$$T^a = De^a = de^a + \omega^a_b \wedge e^b. \quad (456)$$

Metric compatibility and orientation preservation imply

$$D\star = \star D \quad (457)$$

on internal Lorentz tensors.

C.4 Hodge pairing identity

For internal bivectors X, Y ,

$$(\star X)_{ab} Y^{ab} = X_{ab} (\star Y)^{ab} \quad (458)$$

The same identity holds when X and Y carry differential-form degrees, with the exterior product understood:

$$(\star X)_{ab} \wedge Y^{ab} = X_{ab} \wedge (\star Y)^{ab}. \quad (459)$$

C.5 Einstein–Holst normalization

We adopt

$$S_g[e, \omega] = \frac{1}{2\kappa} \int_M B^{ab} \wedge \left[(\star R)_{ab} + \frac{1}{\gamma} R_{ab} \right] - \frac{\Lambda}{\kappa} \int_M \eta \quad (460)$$

With

$$\kappa = \frac{8\pi G}{c^4}, \quad (461)$$

this is the Einstein–Holst normalization used in the present paper, up to the overall sign associated with the curvature convention.

If the action is written in excitation form as

$$S_g = \frac{1}{2} \int_M H_{ab} \wedge R^{ab} - \frac{\Lambda}{\kappa} \int_M \eta, \quad (462)$$

then comparison with Eq. (460) requires

$$H = \frac{1}{\kappa} \left(\star + \frac{1}{\gamma} I \right) B \quad (463)$$

Thus, in the notation

$$\mathfrak{C} = \alpha I + \beta \star, \quad (464)$$

the standard normalization used here is

$$\alpha = \frac{1}{\kappa\gamma}, \quad \beta = \frac{1}{\kappa} \quad (465)$$

If instead one chooses

$$H = \frac{1}{2\kappa} \left(\star + \frac{1}{\gamma} I \right) B$$

while retaining the prefactor $\frac{1}{2} \int H \wedge R$, the entire curvature action is smaller by a factor of two. Such a convention is mathematically consistent, but the symbol κ should then not simultaneously be identified with the standard Einstein normalization without compensating for this factor.

C.6 Matter-current conventions

We define the coframe and Lorentz-spin currents by

$$\delta S_m = \int_M \delta e^a \wedge \Sigma_a - \frac{1}{2} \int_M \delta \omega^{ab} \wedge \tau_{ab} + \text{matter equations} + \text{boundary terms} \quad (466)$$

Accordingly,

$$\tau_{ab} = -2 \frac{\delta S_m}{\delta \omega^{ab}} \quad (467)$$

with the variational ordering used here.

For

$$S_g = \frac{1}{2} \int H_{ab} \wedge R^{ab},$$

the connection variation gives, after discarding a boundary term,

$$\begin{aligned} \delta_\omega S_g &= \frac{1}{2} \int H_{ab} \wedge D \delta \omega^{ab} \\ &= -\frac{1}{2} \int D H_{ab} \wedge \delta \omega^{ab} \\ &= \frac{1}{2} \int \delta \omega^{ab} \wedge D H_{ab}. \end{aligned} \quad (468)$$

Combining this with Eq. (466) gives

$$D H_{ab} = \tau_{ab} \quad (469)$$

C.7 Nieh–Yan identity

Using the first Bianchi identity

$$D T^a = R^a_b \wedge e^b, \quad (470)$$

one finds

$$\begin{aligned} d(e_a \wedge T^a) &= D(e_a \wedge T^a) \\ &= D e_a \wedge T^a - e_a \wedge D T^a \\ &= T_a \wedge T^a - e_a \wedge R^a_b \wedge e^b. \end{aligned} \quad (471)$$

Hence

$$\boxed{d(e_a \wedge T^a) = T_a \wedge T^a - e_a \wedge e_b \wedge R^{ab}} \quad (472)$$

In particular,

$$T^a = 0 \implies e_a \wedge e_b \wedge R^{ab} = 0. \quad (473)$$

D Algebraic Spin–Torsion Response

This appendix proves the invertibility of the algebraic map

$$\mathcal{T}_e(T)^{ab} = T^a \wedge e^b - e^a \wedge T^b$$

and gives an explicit inverse. This permits the spin–torsion response to be written without leaving the inverse map $\mathcal{T}_e^{-1}$ formal.

Let

$$\{e^a\}$$

be a nondegenerate coframe and let

$$\{E_a\}$$

denote its dual frame:

$$\iota_a e^b := \iota_{E_a} e^b = \delta_a^b. \quad (474)$$

For a vector-valued torsion two-form

$$T^a \in \Omega^2(M),$$

define

$$\boxed{K^{ab} := \mathcal{T}_e(T)^{ab} = T^a \wedge e^b - e^a \wedge T^b} \quad (475)$$

The object K^{ab} is antisymmetric in the internal indices and is a three-form.

D.1 Explicit inverse of $\mathcal{T}_e$

Define the torsion trace one-form

$$t := \iota_a T^a. \quad (476)$$

Contract Eq. (475) with ι_b , summing over b :

$$\iota_b K^{ab} = \iota_b (T^a \wedge e^b) - \iota_b (e^a \wedge T^b). \quad (477)$$

For any two-form X ,

$$\sum_b e^b \wedge \iota_b X = 2X, \quad (478)$$

and hence

$$\sum_b (\iota_b X) \wedge e^b = -2X. \quad (479)$$

Therefore

$$\begin{aligned}\iota_b(T^a \wedge e^b) &= (\iota_b T^a) \wedge e^b + T^a \iota_b e^b \\ &= -2T^a + 4T^a \\ &= 2T^a.\end{aligned}\tag{480}$$

Similarly,

$$\begin{aligned}\iota_b(e^a \wedge T^b) &= \delta_b^a T^b - e^a \wedge \iota_b T^b \\ &= T^a - e^a \wedge t.\end{aligned}\tag{481}$$

Hence

$$\boxed{\iota_b K^{ab} = T^a + e^a \wedge t}\tag{482}$$

Contract once more:

$$\begin{aligned}\iota_a \iota_b K^{ab} &= \iota_a T^a + \iota_a(e^a \wedge t) \\ &= t + 4t - e^a \wedge \iota_a t \\ &= t + 4t - t \\ &= 4t.\end{aligned}\tag{483}$$

Thus

$$\boxed{t = \frac{1}{4} \iota_a \iota_b K^{ab}}\tag{484}$$

Substitution into Eq. (482) gives the desired inverse:

Theorem D.1 (Explicit inverse of the torsion map). *For a nondegenerate four-dimensional coframe,*

$$\boxed{\left[\mathcal{T}_e^{-1}(K)\right]^a = \iota_b K^{ab} - \frac{1}{4} e^a \wedge \iota_c \iota_d K^{cd}}\tag{485}$$

Consequently,

$$\mathcal{T}_e : \Omega^2(M, V) \longrightarrow \Omega^3(M, \Lambda^2 V)$$

is an algebraic isomorphism.

Proof. Equation (485) was derived directly from $K = \mathcal{T}_e(T)$, and reconstructs T uniquely from K . It is therefore both a left and right inverse on the finite-dimensional fiber spaces. $\square$

D.2 Explicit constitutive spin–torsion formula

Let

$$\mathfrak{C} = \alpha I + \beta \star, \quad \alpha^2 + \beta^2 \neq 0.\tag{486}$$

The connection equation is

$$\mathfrak{C} \mathcal{T}_e(T) = \tau.\tag{487}$$

Since

$$\mathfrak{C}^{-1} = \frac{\alpha I - \beta \star}{\alpha^2 + \beta^2},\tag{488}$$

define

$$K^{ab} := [\mathfrak{C}^{-1}\tau]^{ab}. \quad (489)$$

Then

$$T^a = \iota_b [\mathfrak{C}^{-1}\tau]^{ab} - \frac{1}{4} e^a \wedge \iota_c \iota_d [\mathfrak{C}^{-1}\tau]^{cd} \quad (490)$$

This is the explicit form of the algebraic response written schematically in the main text as

$$T = \mathcal{T}_e^{-1} \mathfrak{C}^{-1} \tau.$$

D.3 Irreducible torsion data

It is often useful to separate the torsion into its Lorentz-irreducible trace, axial, and tensor pieces. Define

$$t := \iota_a T^a, \quad (491)$$

$$a := \star_{\text{sp}}(e_a \wedge T^a), \quad (492)$$

where $\star_{\text{sp}}$ denotes the spacetime Hodge operator induced by the soldered Lorentz metric.

The remaining tensor piece q^a is characterized by

$$\iota_a q^a = 0, \quad e_a \wedge q^a = 0. \quad (493)$$

Thus the 24 torsion components decompose into

$$24 = 4_{\text{trace}} + 4_{\text{axial}} + 16_{\text{tensor}} \quad (494)$$

The minimally coupled Dirac field occupies only the axial source sector before the Holst constitutive operator is inverted. A nonzero identity component of $\mathfrak{C}$ may mix the corresponding algebraic torsion structures.

D.4 Dirac source in constitutive form

Let

$$J_5 = J_{5a} e^a \quad (495)$$

be the fermionic axial-current one-form. Introduce

$$X_{ab} := J_{5a} \eta_b - J_{5b} \eta_a. \quad (496)$$

Then

$$(\star X)_{ab} = \epsilon_{ab}{}^{cd} J_{5c} \eta_d. \quad (497)$$

We write the minimally coupled Dirac spin current as

$$\tau^{(\text{D})} = s_{\text{D}} \star X \quad (498)$$

where the numerical constant s_{D} depends only on the normalization and γ_5 conventions of the Dirac action.

For the common convention in which

$$\tau_{ab}^{(D)} = \frac{1}{2} \epsilon_{ab}{}^{cd} J_{5c} \eta_d, \quad (499)$$

one has

$$s_D = \frac{1}{2}. \quad (500)$$

Applying the inverse constitutive map gives

$$\begin{aligned} \mathfrak{C}^{-1} \tau^{(D)} &= \frac{s_D}{\alpha^2 + \beta^2} (\alpha I - \beta \star) \star X \\ &= \frac{s_D}{\alpha^2 + \beta^2} (\alpha \star X + \beta X). \end{aligned} \quad (501)$$

The following identities are useful:

$$\boxed{\mathcal{T}_e^{-1}(X)^a = -J_5^b \iota_b \eta^a} \quad (502)$$

$$\boxed{\mathcal{T}_e^{-1}(\star X)^a = -\frac{1}{2} e^a \wedge J_5} \quad (503)$$

Using

$$\iota_b \eta^a = \frac{1}{2} \epsilon^a{}_{bcd} e^c \wedge e^d, \quad (504)$$

one obtains

Proposition D.2 (Dirac torsion for the general constitutive map). *For the Dirac source (498),*

$$\boxed{T_D^a = -\frac{s_D}{2(\alpha^2 + \beta^2)} \left[\beta \epsilon^a{}_{bcd} J_5^b e^c \wedge e^d + \alpha e^a \wedge J_5 \right]} \quad (505)$$

Thus the Hodge component produces the familiar axial torsion channel, whereas the identity/Holst component generates the second algebraic structure proportional to

$$e^a \wedge J_5.$$

D.5 Einstein–Holst parameterization

Using the normalization of Appendix C,

$$\alpha = \frac{1}{\kappa \gamma}, \quad \beta = \frac{1}{\kappa}. \quad (506)$$

Hence

$$\frac{\beta}{\alpha^2 + \beta^2} = \kappa \frac{\gamma^2}{1 + \gamma^2}, \quad (507)$$

$$\frac{\alpha}{\alpha^2 + \beta^2} = \kappa \frac{\gamma}{1 + \gamma^2}. \quad (508)$$

Equation (505) becomes

$$T_D^a = -\frac{s_D \kappa}{2} \frac{\gamma^2}{1 + \gamma^2} \left[\epsilon^a{}_{bcd} J_5^b e^c \wedge e^d + \frac{1}{\gamma} e^a \wedge J_5 \right] \quad (509)$$

In the Einstein–Cartan limit

$$|\gamma| \longrightarrow \infty,$$

this reduces to

$$T_{EC}^a = -\frac{s_D \kappa}{2} \epsilon^a{}_{bcd} J_5^b e^c \wedge e^d \quad (510)$$

With $s_D = 1/2$,

$$T_{EC}^a = -\frac{\kappa}{4} \epsilon^a{}_{bcd} J_5^b e^c \wedge e^d, \quad (511)$$

up to the sign determined by the curvature, γ_5 , and matter-variation conventions.

D.6 Response-loop phase

Insertion into the infinitesimal response-loop theorem gives

$$\log W_{\partial\Sigma}(P) = \frac{is_D}{2\hbar(\alpha^2 + \beta^2)} \int_{\Sigma} P_a \left[\beta \epsilon^a{}_{bcd} J_5^b e^c \wedge e^d + \alpha e^a \wedge J_5 \right] + O(\ell^3) \quad (512)$$

This formula exhibits directly the operational chain

$$J_5 \longrightarrow \tau \longrightarrow T \longrightarrow \arg W_{\partial\Sigma} \quad (513)$$

E First-Order Variations and the Spinless Einstein Equivalence Class

This appendix derives the connection and coframe equations for the general constitutive action and shows directly, at the level of field equations, why the identity/Holst direction disappears in the torsion-free spinless sector.

Consider

$$S[e, \omega] = \frac{1}{2} \int_M \mathfrak{C}(B)_{ab} \wedge R^{ab} - \frac{\Lambda}{\kappa} \int_M \eta + S_m, \quad (514)$$

where

$$B^{ab} = e^a \wedge e^b \quad (515)$$

and

$$\mathfrak{C} = \alpha I + \beta \star. \quad (516)$$

The coefficients are assumed constant in the restricted constitutive sector.

E.1 Connection variation

Since B is held fixed under an independent connection variation,

$$\delta_\omega B^{ab} = 0.$$

Furthermore,

$$\delta_\omega R^{ab} = D\delta\omega^{ab}.$$

Therefore

$$\begin{aligned}\delta_\omega S_g &= \frac{1}{2} \int H_{ab} \wedge D\delta\omega^{ab} \\ &= \frac{1}{2} \int \delta\omega^{ab} \wedge DH_{ab} + \text{boundary}.\end{aligned}\tag{517}$$

With

$$\delta_\omega S_m = -\frac{1}{2} \int \delta\omega^{ab} \wedge \tau_{ab},$$

the field equation is

$$\boxed{DH_{ab} = \tau_{ab}}\tag{518}$$

For constant α, β ,

$$D\mathfrak{C} = 0,$$

so

$$\mathfrak{C}(DB) = \tau.\tag{519}$$

Using

$$DB = \mathcal{T}_e(T),$$

one recovers

$$\boxed{\mathfrak{C}\mathcal{T}_e(T) = \tau}\tag{520}$$

If

$$\tau = 0$$

and

$$(\alpha, \beta) \neq (0, 0),$$

both $\mathfrak{C}$ and $\mathcal{T}_e$ are invertible, and hence

$$\boxed{T^a = 0}\tag{521}$$

E.2 Coframe variation

We next vary the coframe while holding the connection fixed. Since

$$B^{ab} = e^a \wedge e^b,\tag{522}$$

$$\begin{aligned}\delta B^{ab} &= \delta e^a \wedge e^b + e^a \wedge \delta e^b \\ &= 2\delta e^{[a} \wedge e^{b]}.\end{aligned}\tag{523}$$

Because I and $\star$ are self-adjoint with respect to the internal bivector pairing,

$$\mathfrak{C}(\delta B)_{ab} \wedge R^{ab} = \delta B_{ab} \wedge \mathfrak{C}(R)^{ab}. \quad (524)$$

Therefore

$$\begin{aligned} \delta_e S_g^{(\text{curv})} &= \frac{1}{2} \int \delta B_{ab} \wedge \mathfrak{C}(R)^{ab} \\ &= \int \delta e^a \wedge e^b \wedge \mathfrak{C}(R)_{ab}. \end{aligned} \quad (525)$$

The volume variation is

$$\boxed{\delta \eta = \delta e^a \wedge \eta_a} \quad (526)$$

Thus, with

$$\delta_e S_m = \int \delta e^a \wedge \Sigma_a,$$

the coframe equation is

$$\boxed{e^b \wedge \mathfrak{C}(R)_{ab} - \frac{\Lambda}{\kappa} \eta_a + \Sigma_a = 0} \quad (527)$$

with the matter sign convention adopted here.

E.3 Torsion-free reduction

Substituting

$$\mathfrak{C} = \alpha I + \beta \star$$

gives

$$e^b \wedge \mathfrak{C}(R)_{ab} = \alpha e^b \wedge R_{ab} + \beta e^b \wedge (\star R)_{ab}. \quad (528)$$

The first Bianchi identity is

$$DT^a = R^a_b \wedge e^b. \quad (529)$$

On the torsion-free branch,

$$T^a = 0,$$

and therefore

$$\boxed{R^a_b \wedge e^b = 0} \quad (530)$$

Since a two-form commutes with a one-form under the exterior product, this is equivalently

$$\boxed{e^b \wedge R_{ab} = 0} \quad (531)$$

Hence the entire identity/Holst contribution to the torsion-free coframe equation vanishes:

$$\alpha e^b \wedge R_{ab} = 0. \quad (532)$$

The remaining field equation is

$$\boxed{\beta e^b \wedge (\star R)_{ab} - \frac{\Lambda}{\kappa} \eta_a + \Sigma_a = 0} \quad (533)$$

For

$$\beta \neq 0,$$

this is the Einstein equation in coframe form, with its overall matter–curvature normalization determined by β .

E.4 Field-equation version of the equivalence theorem

Theorem E.1 (Torsion-free constitutive equivalence). *Consider the family*

$$\mathfrak{C}_{\alpha,\beta} = \alpha I + \beta \star, \quad \beta \neq 0.$$

In the spinless nondegenerate algebraic connection branch,

$$\tau = 0$$

implies

$$T = 0.$$

On that branch, the coframe field equation is independent of α . Therefore changing α at fixed nonzero β does not change the classical torsion-free bulk field equations.

Proof. The connection equation and invertibility of $\mathfrak{C}\mathcal{T}_e$ give

$$T = 0.$$

The first Bianchi identity then gives

$$e^b \wedge R_{ab} = 0.$$

Consequently the αI component of $\mathfrak{C}(R)$ drops out of the coframe equation identically. Only the $\beta \star$ term remains. $\square$

E.5 Relation to the Nieh–Yan argument

The density identity

$$T = 0 \implies B_{ab} \wedge R^{ab} = 0$$

and the field-equation identity

$$T = 0 \implies e^b \wedge R_{ab} = 0$$

are related but logically distinct.

The first proves that the Holst density vanishes on the torsion-free branch. The second proves directly that its coframe variation contributes no independent torsion-free field equation.

For the equivalence theorem in the main text, the second form is the stronger statement.

F Further Properties of Constitutive Reciprocity

This appendix collects several algebraic consequences and limitations of the constitutive reciprocity principle introduced in Sec. 7.

F.1 Graph invariance is equivalent to closure

Let

$$H = \mathbb{C}B$$

and define

$$\mathcal{R}_\zeta(B, H) = (\zeta H, -\zeta^{-1}B), \quad \zeta \neq 0. \quad (534)$$

The main text proves that invariance of the constitutive graph implies

$$\mathbb{C}^2 = -\zeta^{-2}I.$$

The converse also holds.

Proposition F.1 (Closure–reciprocity equivalence). *For a linear constitutive map $\mathbb{C}$, the graph*

$$\Gamma_{\mathbb{C}} = \{(B, H) : H = \mathbb{C}B\}$$

is invariant under $\mathcal{R}_\zeta$ if and only if

$$\boxed{\mathbb{C}^2 = -\zeta^{-2}I} \quad (535)$$

Proof. The forward implication was proved in the main text.

Conversely, suppose

$$\mathbb{C}^2 = -\zeta^{-2}I$$

and let

$$H = \mathbb{C}B.$$

After reciprocity,

$$B' = \zeta \mathbb{C}B, \quad H' = -\zeta^{-1}B.$$

Then

$$\mathbb{C}B' = \zeta \mathbb{C}^2 B = -\zeta^{-1}B = H'.$$

Thus

$$(B', H') \in \Gamma_{\mathbb{C}}.$$

□

F.2 Complex-structure form

Define

$$J = \zeta \mathbb{C}. \quad (536)$$

Then closure is simply

$$\boxed{J^2 = -I} \quad (537)$$

Thus the reciprocity-closed constitutive operator equips the real bivector space with a complex structure.

Lorentz equivariance further requires

$$J = aI + b \star.$$

The condition

$$J^2 = -I$$

then yields

$$2ab = 0, \quad a^2 - b^2 = -1.$$

For real a, b , the only solutions are

$$\boxed{J = \pm \star} \tag{538}$$

F.3 Euclidean comparison

Suppose instead that the internal metric were Euclidean in four dimensions. Then

$$\star^2 = +I$$

on real two-forms. A real equivariant map

$$\mathfrak{C} = \alpha I + \beta \star$$

would satisfy

$$\mathfrak{C}^2 = (\alpha^2 + \beta^2)I + 2\alpha\beta \star. \tag{539}$$

Pure reciprocity closure would require

$$2\alpha\beta = 0, \tag{540}$$

$$\alpha^2 + \beta^2 = -\zeta^{-2}. \tag{541}$$

The second equation has no real solution.

Therefore

$$\boxed{\text{real pure reciprocity closure is compatible with the Lorentzian bivector complex structure but not with the Euclidean}} \tag{542}$$

In the present reconstruction this is a consistency result rather than a derivation of spacetime signature, because Lorentzian signature has already been reconstructed before the constitutive sector is introduced.

F.4 Generalized shifted closure

Consider instead

$$(\mathfrak{C} - \alpha_0 I)^2 = -\lambda^2 I. \tag{543}$$

Then

$$\boxed{\mathfrak{C} = \alpha_0 I \pm \lambda \star} \tag{544}$$

is a solution.

Thus a shifted closure relation retains an arbitrary identity component:

$$\boxed{\text{shifted closure} \not\Rightarrow \text{absence of the Holst direction}} \tag{545}$$

The selection

$$\mathfrak{C} \propto \star$$

therefore depends specifically on the pure reciprocity transformation used in the main text.

F.5 Scale freedom

Suppose

$$\mathfrak{G}^2 = -\zeta^{-2}I.$$

For any positive number λ , define

$$\mathfrak{G}' = \lambda\mathfrak{G}, \quad \zeta' = \frac{\zeta}{\lambda}. \quad (546)$$

Then

$$\begin{aligned} (\mathfrak{G}')^2 &= \lambda^2\mathfrak{G}^2 \\ &= -\lambda^2\zeta^{-2}I \\ &= -(\zeta')^{-2}I. \end{aligned} \quad (547)$$

Thus reciprocity closure is invariant under the simultaneous rescaling

$$\boxed{(\mathfrak{G}, \zeta) \mapsto \left(\lambda\mathfrak{G}, \frac{\zeta}{\lambda}\right)} \quad (548)$$

Consequently the closure condition cannot determine the overall constitutive magnitude.

For the Einstein direction,

$$\mathfrak{G} = \frac{1}{\kappa}\star,$$

this is precisely the statement that

$$\kappa$$

and hence the numerical value of G remain independent constitutive calibration data.

F.6 Orientation reversal

The Hodge operator depends on the chosen internal orientation. Under orientation reversal,

$$\star \mapsto -\star. \quad (549)$$

Hence the two reciprocity solutions

$$\mathfrak{G} = \pm\zeta^{-1}\star$$

are exchanged.

The remaining sign should therefore not be interpreted as an independent continuous gravitational coupling. Its physical status depends on the chosen orientation and overall sign convention for the first-order action.

G Local Response Reversibility Does Not Determine the Constitutive Map

This appendix makes precise why the constitutive reciprocity principle of Sec. 7 cannot be inferred merely from reversibility or unitarity of the local operational response.

G.1 Local response reversal

The local phase associated with the generator–displacement response is

$$W(P, \xi) = \exp \left[-\frac{i}{\hbar} \Theta(P, \xi) \right]. \quad (550)$$

Since

$$\Theta(P, \xi)$$

is real,

$$\boxed{W(P, \xi)^\dagger = W(P, \xi)^{-1}} \quad (551)$$

When the response is linear,

$$W(P, \xi)^{-1} = W(-P, \xi) = W(P, -\xi). \quad (552)$$

Thus local response reversal is an involution:

$$\boxed{\mathcal{R}_W^2 = I} \quad (553)$$

G.2 Constitutive reciprocity has different algebraic type

By contrast, the area–excitation reciprocity map is

$$\mathcal{R}_\zeta(B, H) = (\zeta H, -\zeta^{-1} B), \quad (554)$$

and satisfies

$$\boxed{\mathcal{R}_\zeta^2 = -I, \quad \mathcal{R}_\zeta^4 = I} \quad (555)$$

The two operations therefore have different algebraic character:

$$\boxed{\text{local response reversal is involutive, whereas constitutive reciprocity is a quarter-turn}} \quad (556)$$

Unitarity alone supplies no canonical operation converting the former into the latter.

G.3 The missing excitation variable

More fundamentally, the local response

$$\Theta(P, \xi) = P(e\xi)$$

determines

$$e, \quad g, \quad B = e \wedge e,$$

but contains no independent curvature excitation H .

The constitutive reciprocity map acts on the doubled space

$$(B, H).$$

Therefore the very statement

$$B \mapsto H$$

already presupposes information not contained in the local response alone.

This observation can be converted into a constructive underdetermination result.

Proposition G.1 (Local-response constitutive no-go). *Fix the operational data*

$$\mathfrak{A}_{\text{local}} = \{\Theta, e, g, B, W\}. \quad (557)$$

For every pair

$$(\alpha, \beta) \neq (0, 0),$$

define

$$\mathfrak{C}_{\alpha, \beta} = \alpha I + \beta \star. \quad (558)$$

Then all members of this continuous family share exactly the same local operational data $\mathfrak{A}_{\text{local}}$ while defining distinct curvature excitations

$$H_{\alpha, \beta} = \mathfrak{C}_{\alpha, \beta}(B). \quad (559)$$

Consequently,

$$\boxed{\Theta, e, g, B, W, W^\dagger \not\Rightarrow \mathfrak{C}} \quad (560)$$

Proof. None of

$$\Theta(P, \xi) = P(e\xi), \quad g = e^* \eta, \quad B = e \wedge e, \quad W = e^{-i\Theta/\hbar}$$

contains the coefficients α, β . Changing

$$(\alpha, \beta)$$

therefore changes H without changing any of these operational objects.

Since two distinct constitutive maps are compatible with identical local response data, no functional of those local response data alone can uniquely determine the constitutive map. $\square$

G.4 Failure of a derivation of pure reciprocity

Pure reciprocity closure requires

$$\mathfrak{C}^2 = -\zeta^{-2} I. \quad (561)$$

The family

$$\mathfrak{C}_{\alpha, \beta} = \alpha I + \beta \star$$

contains arbitrarily many invertible maps that do not satisfy this condition.

For example,

$$\mathfrak{C} = I \quad (562)$$

is real, local, invertible, and Lorentz equivariant, and may be placed on exactly the same operationally reconstructed geometry. Nevertheless,

$$\mathfrak{C}^2 = I$$

cannot equal

$$-\zeta^{-2} I$$

for real ζ .

Therefore

$$\boxed{\text{local response composition and reversibility} \not\Rightarrow \text{pure constitutive reciprocity}} \quad (563)$$

The reciprocity principle used in Sec. 7 is consequently an independent constitutive assumption.

G.5 Logical boundary

The result of this appendix may be summarized as

$$\begin{array}{l}
 \Theta \implies e, g, B, \\
 W^\dagger = W^{-1} \implies \text{response reversibility}, \\
 \Theta, W \not\Rightarrow H, \\
 \Theta, W \not\Rightarrow \mathfrak{C}, \\
 \Theta, W \not\Rightarrow \mathcal{R}_\zeta.
 \end{array} \tag{564}$$

This is the precise sense in which constitutive reciprocity lies beyond the operational kinematics reconstructed from the local response.